

CHEMISTRY

ATOMIC STRUCTURE, RADIOACTIVITY, PERIODIC TABLE

1. The nucleus of an atom consists of [67Re BPSC]

- A. Electrons and protons
- B. Protons and neutrons
- C. Electrons only
- D. Electrons and neutrons
- E. None of the above /more than one of the above

Ans. B

Exp: Atoms are electrically neutral because they contain equal quantities of positively charged protons and negatively charged electrons. The nucleus contains protons and neutrons, and the electrons occupy a “cloud” **around the outside of the nucleus inside an atom**. The **mass of an electron** is only about 1/2000 the mass of a proton or neutron, so electrons contribute virtually nothing to the total mass of an atom (**negligible**). The unit of measure for mass is the atomic mass unit (AMU), and one atomic mass unit (1amu) is equal to 1.66×10^{-24} grams. The nucleus consists of protons and neutrons bound by a strong nuclear force. It was discovered by Ernest Rutherford.

BCW Bits:

Name	Symbol Name	Absolute Charge /C	Relative Charge	Mass/kg	Discovered
Electron	e	$-1.60217663 \times 10^{-19}$	-1	$9.1093837 \times 10^{-31}$	J.J Thomson
Proton	p	$+1.60217663 \times 10^{-19}$	+1	$1.6726216 \times 10^{-27}$	Ernest Rutherford
Neutron	n	0	0	$1.6726216 \times 10^{-27}$	James Chadwick

2. Which of the following is an incorrect option? [67 Cancelled BPSC]

- A. Atom is electrically neutral due to the presence of protons and electrons inside the nucleus
- B. Electrons are negatively charged and have negligible mass
- C. Atomic number is the number of protons in an atom of an element.
- D. Isotopes are useful as nuclear fuel, in the medical field, in carbon dating, etc.
- E. None of the above/more than one of the above

Ans. A

Exp: Refer to Que No 1

3. The atomic nucleus was discovered by [29th Bihar Judicial Service]

- A. Rutherford
- B. Dalton
- C. Einstein
- D. Thompson

Ans. A

Exp: The atomic nucleus was discovered by Rutherford, through his famous alpha-particle gold foil experiment.

BCW Bits:

- ❖ He wanted to find out the arrangement of electrons within an atom, so he designed an experiment in which fast-moving alpha particles (doubly charged helium atoms) were made to fall on a thin gold foil.
- ❖ He expected that alpha particles would be deflected by the subatomic particles in the gold atoms. But large deflections were not expected as alpha particles were much heavier than the protons. Most of the alpha particles passed straight from the gold foil, which means most of the space in the atom is empty.
- ❖ Some of them were deflected by small angles indicating that the positive charge of the atom occupies very little space and surprisingly one out of every 12,000 particles appeared to rebound (deflected by 180 degrees) indicating the concentration of all the positive charge and mass of the gold foil atom in a very small volume within the atom called the Nucleus.
- ❖ Thus, the atomic nucleus was discovered and he also calculated that the radius of the nucleus is about 10^5 times less than the radius of the atom.

4. Isotopes of an element differ in [67 Cancelled BPSC]

- A. The number of protons
- B. The mass numbers
- C. The number of electrons
- D. The atomic number
- E. None of the above/more than one of the above

Ans. B

Exp: **Isotopes** are atoms of the same element that have different numbers of neutrons but the same number of protons and electrons. The **difference in the number of neutrons** between the various isotopes of an element means that the various isotopes have **different mass numbers**. The mass number is the sum of protons and neutrons in the nucleus, and is denoted by 'A'.

BCW Bits:

- ❖ Isobars are atoms of different chemical elements having the same number of nucleons (mass number) but different atomic numbers (Z).

- ❖ Isotopes are atoms of chemical elements that have the same atomic number but a different atomic mass.
- ❖ Isotones are atoms with the same numbers of neutrons but different proton numbers.

Name	Definition	Example
Isotopes	Same Z, Different A	${}_8\text{O}^{15}$, ${}_8\text{O}^{16}$, ${}_8\text{O}^{17}$
Isobars	Same A, Different Z	${}_{18}\text{Ar}^{40}$, ${}_{19}\text{K}^{40}$, Ca^{40}
Isotones	Same N, Different Z	${}_{15}\text{P}^{31}$, ${}_{16}\text{S}^{32}$

Where,

A=Atomic mass**Z**=Atomic number**N**=Number of Neutrons

Hydrogen does not contain neutrons, because its nucleus is the smallest in size. It also makes hydrogen atoms unstable.

There are three isotopes of hydrogen namely, protium, deuterium and tritium.

Deuterium when reacts with oxygen forms D_2O . This D_2O is one of the best moderators in Nuclear Plants. It is to be noted that the moderator reduces the speed of the neutron and hence helps in controlling the Nuclear Fission Reaction.

5. The mass number of a nucleus is [67 Cancelled BPSC]

- A. The sum of the number of neutrons and protons
- B. The total mass of neutrons and protons
- C. Always more than the atomic weight
- D. Always less than its atomic number
- E. None of the above/more than one of the above

Ans. A**Exp:** Refer to Que No 4**6. The Mass Number of an element is equal to the number of ____ present in an atom. [District Art and Culture BPSC]**

- A. Protons
- B. Fermions
- C. Electrons
- D. Protons And Neutrons

- E. None of the above/more than one of the above.

Ans. D

Exp: Refer to Que No 4

7. The number of neutrons in the nucleus of plutonium nuclide (${}_{94}\text{Pu}^{242}$) is [66th BPSC]

- A. 94 B. 148
C. 242 D. 336

- E. None of the above/More than one of the above

Ans. B

Exp: Refer to Que No 4

- ❖ No of Neutrons = Mass no of Plutonium - No of Protons (atomic number) = $242 - 94 = 148$ Plutonium:
- ❖ It belongs to the Element group: Actinides in the periodic table.
- ❖ Arthur C. Wohl, Glenn T. Seaborg, and Joseph W. Kennedy in 1940 produced it for the first time.

8. The number of electrons and neutrons in an element is 18 and 20, respectively. Its mass number is [65th BPSC]

- A. 22 B. 2
C. 38 D. 20

- E. None of the above/More than one of the above

Ans. C

Exp: Refer to Que No 4

9. The positively charged part at the centre of an atom is called as [63rd BPSC]

- A. Proton B. Neutron
C. Electron D. Nucleus

- E. None of these/More than one of these

Ans. D

Exp: Atoms are composed of three elementary particles, electron, proton, and neutron. Neutrons and protons are present in the nucleus. Due to the presence of positively charged protons and electrically neutral neutrons, the nucleus is positively charged.

10. Which of the following elements does not contain neutrons? [64th BPSC]

- A. Oxygen B. Nitrogen
C. Hydrogen D. Copper

- E. None of the above/More than one of the above

Ans. C

Exp: Refer to Que No 4

11. Nucleus of helium contains [29th Bihar Judicial Service]

- A. Only one proton
B. Two protons
C. Two protons and two neutrons
D. One proton and two neutrons

Ans. C

Exp: Nucleus of helium contains two protons and two neutrons.

BCW Bits:

Every element is represented as ${}_Z^AX$, where Z denotes the atomic number (which is equal to the no of electrons (it is present outside the nucleus) or no of protons inside the nucleus and A denotes the atomic mass (which is equal to the sum of the number of protons and neutrons present in the nucleus of an atom). The atomic number of Helium is '2' i.e., no of electrons=no. of protons=2, and atomic mass is '4' so the number of neutrons is equal to $A - Z$ (as $A = Z$ (proton) + N (neutron)) i.e., $4 - 2 = 2$.

Hence, the nucleus of a helium atom has two protons and two neutrons.

- ❖ Helium (He) is a group-18 element. It belongs to the inert or noble gas category of the periodic table. It is the second lightest element after hydrogen.

12. Which particle is free of charge? [66 Re BPSC]

- A. Particle B. Electron
C. Neutron D. Proton

- E. None of the above/More than one of the above

Ans. C

Exp: Refer to Que No 1

13. The modern atomic symbols are based on the method proposed by [CDPO 2022 BPSC]

- A. Bohr B. Berzelius
C. Dalton D. Thomson
E. None of the above/More than one of the above

Ans. B

Exp: Jons Jacob Berzelius suggested a new system of nomenclature by using one or two letters of elements. Example: He symbolized carbon as 'C', oxygen as 'O', and compounds like water were represented as H_2O , denoting that it contains 2 atoms of hydrogen and one atom of oxygen. So, he revolutionized the chemistry nomenclature system.

14. Who was the first to suggest that matter is made up of tiny particles called Parmanu [District Art and Culture BPSC]

- A. Maharishi Kanad
B. John Dalton
C. Robert Brown
D. Ernest Rutherford
E. None of the above/more than one of the above.

Ans. A

Exp: Maharishi Kanad was the first one to give a theory on how atoms combine to form molecules and compounds, in his book **Vaisheshika Sutra**.

15. The structural stability of a matter is explained first time by [District Art and Culture BPSC]

- A. Ernest Rutherford
B. Niels Bohr
C. James Chadwick
D. J.J Thomson
E. None of the above/more than one of the above.

Ans. B.

Exp: Bohr proposed stability through the concept of the revolution of electrons in different energy levels. The change in the energy level of an electron occurs when it jumps from lower energy to higher energy levels and vice versa, electrons move from higher energy to a lower energy level

by losing energy and electrons move from lower energy to higher energy level by gaining the required energy.

J.J Thomson -J.J. Thomson was the first one to propose a model for the structure of an atom, known as **plum-pudding model**.

16. Who is regarded as the Father of Modern Chemistry? [65th BPSC]

- A. Rutherford B. Einstein
C. Lavoisier D. C V Raman
E. None of the above/More than one of the above

Ans. C

Exp: **Antoine Lavoisier** is known as the **father of modern chemistry**. He named the elements carbon, hydrogen and oxygen and discovered oxygen's role in combustion and respiration.

Ernest Rutherford- He was known as the **'Father' of nuclear physics**. He is famous for his work on radioactivity and the discovery of the nucleus of an atom with the gold foil experiment.

17. The maximum number of electrons to be accommodated in a shell depends on the formula [Sanitary Waste Management BPSC]

- A. n^2 B. $2n$
C. $3n$ D. $2n^2$

Ans. D

Exp: According to **Pauli's exclusion principle**, the number of electrons that can be accommodated in a shell is given by $2n^2$.

Here 'n' is the orbit number;

E.g. if $n=1$, then, no of electrons in that orbit = $2 \times (1)^2 = 2$

If $n=2$, then, no of electrons in that orbit = $2 \times (2)^2 = 8$

If $n=3$, then, no of electrons in that orbit = $2 \times (3)^2 = 18$

Test Your Knowledge

1. Consider the following statements about the Newland Octave series.

- (i) The last element of this series is thorium.

- (ii) Cobalt and Nickel Occupied the same position.
- (iii) There were 60 elements in this series.
- (iv) The series was based on the atomic mass of the elements.

Which of the following statements are correct?

- A. (i) only
- B. (i) and (ii) only
- C. (i), (ii) and (iv) only
- D. More than one of the above
- E. None of the above.

Ans. C

2. Which of the following statements does not form a part of Bohr's model of the hydrogen atom?

- A. The energy of the electrons in the orbit is quantized
- B. The electron in the orbit nearest the nucleus has the lowest energy
- C. The position and velocity of the electrons in the orbit cannot be determined simultaneously
- D. More than one of the above
- E. None of the above

Ans. C

3. Which of the following statements is/are not true about Neutron?

- A. It does not have any net electric charge associated with them.
- B. It is an atomic particle.
- C. The mass of the neutron is slightly greater than the proton.
- D. More than one of the above
- E. None of the above

Ans. B

4. Choose the correct statement about electro negativity

- A. Electro negativity generally increases across a period from left to right
- B. Decreases down a group in the periodic table
- C. Electro negativity generally decreases across a period from left to right

- D. More than one of the above
- E. None of the above

Ans. D (A, B)

PHYSICAL AND CHEMICAL CHANGES, PROPERTIES OF MATTER, MIXTURE AND SOLUTION, ACID, BASE

18. Which among the following is known as quick lime? [67Re BPSC]

- A. CaCO_2
- B. $\text{Ca}(\text{OH})_2$
- C. CaCl_2
- D. CaO
- E. None of the above more/ than one of the above

Ans. D.

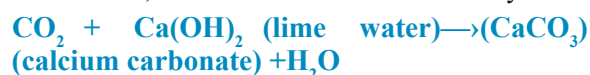
Exp: Calcium Oxide is commonly known as quicklime or burnt lime.

BCW Bits:

- ❖ Calcium oxide reacts vigorously with water to produce slaked lime (calcium hydroxide) releasing large amounts of heat.



- ❖ When carbon dioxide is passed through lime water/slaked lime, calcium carbonate is formed, which makes lime water milky.



19. Limewater is turned milky by [66th BPSC]

- A. CO
- B. CO_2
- C. O_2
- D. O_3
- E. None of the above/ more than one of the above

Ans. B

Exp: Refer to Que No 18

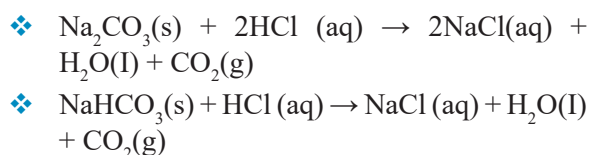
20. Acids react with metal carbonates to liberate _____ gas [68th BPSC Pre]

- A. CO
- B. H_2
- C. CO_2
- D. More than one of the above
- E. None of the above

Ans. C

Exp: Acids react with metal carbonates to liberate Carbon-dioxide (CO_2) gas.

Not only metal carbonates but metal hydrogen carbonates also liberate Carbon-dioxide (CO_2) gas when they react with acids. Some examples are:



21. Consider the following statements [67Re BPSC]

1. Addition of salt to pure water increases the boiling point of water and decreases the freezing point of water.
2. When methyl alcohol is added to water, the boiling point of water decreases.

Which of the above statements is/are correct?

- A. 1 only
- B. 2 only
- C. Both 1 and 2
- D. Salt and methyl alcohol are the same in behaviour
- E. None of the above/ more than one of the above

Ans. C.

Exp: When a volatile solute is added to a solvent, vapour pressure increases which means water evaporates at a lesser temperature which results in a decrease in boiling point.

- ❖ Methyl alcohol when added to water forms a binary solution as both are volatile liquids, as a result, this solution increases the vapour pressure and depression in boiling point.
- ❖ Salt is non-volatile and when dissolves in water it acquires some surface water molecules which will decrease the vapour pressure of water, it will start to boil when the vapour pressure is equal to the atmospheric pressure which means a high temperature is required to boil the water.
- ❖ Other non-volatile solutes are sugar and baking soda.

22. The antifreeze used in automobiles chiefly contains [Bihar Assistant Prosecution Officer]

- A. Water
- B. Acetone
- C. Ethylene Glycol
- D. Ethyl Glycol

Ans. C

Exp: The antifreeze used in automobiles chiefly contains Ethylene Glycol. It is a non-volatile compound, when added to water it decreases the freezing point of the water by reducing the vapour pressure of the solution (water + ethylene glycol) and increases its boiling point hence acting as a coolant as well.

BCW Bits:

Vapour pressure: It can be defined as the measure of the tendency of a substance to change into a gaseous (Vapor) state. It is a pressure exerted by a vapour in thermodynamic equilibrium with its condensed phase (either solid or liquid).

Non-volatile compound: These are the substances when added to a solvent that decreases its freezing point and increases its boiling point. Their tendency is not to vaporize easily.

- ❖ The freezing point of a substance can be defined as the temperature at which the vapour pressure of the substance in its liquid phase is equal to its vapour pressure in the solid phase. A solution will freeze when its vapour pressure becomes equal to the vapour pressure of the pure solid solvent. The addition of a non-volatile solid into the solvent decreases its vapour pressure i.e., it would now become equal to that of the solid solvent at a lower temperature. Thus the freezing point of the solvent decreases.

BCW Concept: As we know that water is a liquid and when it will freeze it will change into a solid state. The particles or atoms of a liquid are less organized when compared to a solid, so to get converted into a solid (when water will freeze) the liquid atom will have to attain a compact crystalline structure (i.e., atoms will have to lose kinetic energy to get more organized). If we add any non-volatile substance to a liquid then it will not be possible for the liquid to get organized at the same temperature at which it used to get organized or frozen earlier because the atoms of the non-volatile substance are also present. It will

be required to reduce the temperature even further to get them frozen. This same thing happens when we add ethylene glycol to water, and hence its freezing point decreases. (i.e., the liquid will freeze at a temperature lower than its actual freezing point for example if it was freezing at 0°C earlier, now it would freeze at -2° or -4°C).

Also, we know that Boiling is a bulk phenomenon. When water boils the water molecules at the surface change into vapor form from the earlier liquid form. And after the addition of any non-volatile substance, the particles of that substance will also be present at the surface. So the amount of water molecules getting converted into vapour form will be less than the earlier value (i.e., vapour pressure would reduce). So more energy or temperature would be required to boil water or get it converted into vapour form. Hence, the boiling point increases with the addition of a non-volatile substance.

23. The temperature point at which solid, liquid and gaseous states may stay together is known as [67 Cancelled BPSC]
- A. Boiling point B. Melting point
C. Freezing point D. Triple point
E. None of the above/more than one of the above

Ans. D

Exp: The **triple point** is the temperature and pressure at which **the solid, liquid, and vapour phases** of a particular substance **co-exist in equilibrium**. It describes a specific thermodynamic state of matter.

The pressure of the gas above a liquid affects the boiling point. The greater the pressure, the more energy is required to boil the liquid and increase the boiling point of the liquid. Higher atmospheric pressure = more energy required to boil = Higher boiling point

24. Water is an excellent cooling agent because of its [CDPO 2022 BPSC]
- A. Low boiling point
B. High boiling point
C. High-specific heat
D. Low specific heat

E. None of the above/More than one of the above

Ans. C

Exp: Water has high specific heat. Water neither heats up nor cools down too quickly; a large amount of energy is absorbed by the water to raise its temperature. Hence, when water is in contact with a body of higher temperature, the water absorbs heat and cools it down. This makes water an excellent cooling agent.

25. Water can dissolve more substances than any other liquid because [UPSC 2021].
- A. It Is Dipolar in Nature
B. It Is a Good Conductor of Heat
C. It Has a High Value of Specific Heat
D. It Is an Oxide of Hydrogen

Ans. A.

Exp: Water can dissolve more substances than any other liquid because of its dipolar nature.

BCW Bits :

Dipolar Nature: A Water molecule consists of two hydrogen atoms and one oxygen atom. Hydrogen atoms present in the water molecule are bonded to the oxygen atom by covalent bonds. A covalent bond is a type of bond in which atoms share electrons between them. In the case of water molecules, the electrons are not equally shared between the hydrogen and oxygen atom. The electrons are more attracted towards the oxygen as the nucleus of the oxygen atom has more protons (8) in comparison to the nucleus of Hydrogen which contains only 1 proton. The larger affinity of covalent bonded electrons towards oxygen is due to the positive charge of protons. This makes the Oxygen atom of the water molecule slightly negative (negative polarity) and the hydrogen atom slightly positive (positive polarity). This also explains the Dipolar nature of the Water molecules and helps it in attracting and dissolving ionic or polar molecules.

Due to this polarity or difference of charge, one water molecule attracts another water molecule through a force of attraction called a '**Hydrogen Bond**'. The hydrogen atom of a water molecule is attracted to the oxygen atom of the adjacent water molecule, thus forming a hydrogen bond

between them. This property helps water to surround the solute with water molecules from all directions. Due to hydrogen bonding with polar molecules, covalent compounds like alcohol and carbohydrates dissolve in water.

- ❖ Hydrogen Bond is a weak bond when compared to an Ionic or Covalent bond because in this bond actual transfer or sharing of electrons does not take place.

Example: when we dissolve NaCl (sodium chloride, also known as table salt) in water. It will dissociate and sodium ions having the positive charge will get attracted to the oxygen atom having the negative polarity and chlorine ions having the negative charge will be attracted towards the positive polarity of the Hydrogen. And as we know water molecules also attract each other through hydrogen bonds, thus the solute gets surrounded by a lot of water molecules as well and it helps the particle to get dissolved in the water evenly.

Hence, option A is the correct answer.

26. With the increase of pressure, the boiling point of the substance [AAO 2022 BPSC]

- A. Increases B. Decreases
C. Remain same D. Become zero
E. None of the above/More than one of the above

Ans. A

Exp: Refer to Que no 23

27. The pH of a solution changes from 3 to 6. The H^+ ion concentration will [66th BPSC]

- A. Increase 3 times
B. Decrease 3 times
C. Decrease 10 times
D. Decrease 1000 times
E. None of the above/More than one of the above

Ans. D

Exp: The pH (potential of hydrogen), is a scale used to specify the acidity or basicity of an aqueous solution. The pH is defined as the negative log of the concentration of H^+ ions. Acidic solutions are measured to have lower pH values than basic or alkaline solutions.

pH value of more than 7 is considered to be basic while less than 7 is considered to be acidic.

A pH of 'a' means concentration of H^+ ions = 10^{-a}

So, Concentration of H^+ ion in pH of 3 = 10^{-3}

And, the concentration of H^+ in pH of 6 = 10^{-6}

pH=3 implies, $pH = -\log_{10}(10^{-3}) = 3$

pH=6 implies, $pH = -\log_{10}(10^{-6}) = 6$

So, it is very clear that H^+ concentration became 10^{-6} from 10^{-3} , i.e. reduction by 1000 times

BCW Bits:

pH:

- ❖ The pH scale was developed by a Danish scientist **Soren Sorensen**.
- ❖ It is a scale for measuring hydrogen ion concentration in a solution.
- ❖ The 'p' in pH stands for 'potenz' in German, meaning power.
- ❖ On the pH scale we can measure pH generally from 0 (very acidic) to 14 (very alkaline).

The pH of common substances:

Gastric Juices	About 1.2
Lemon Juice	About 2.2
Saliva before meal	About 7.4
Saliva after meal	About 5.8
Carrot juice	About 6
Pure water (Neutral Solution)	7
Blood	About 7.4
Milk of magnesia	About 10
Sodium Hydroxide solution	About 14

28. The pH value of water is [66 th Re BPSC]

- A. 4 B. 7
C. 12 D. 18
E. None of the above/More than one of the above

Ans. B

Exp: Refer to Que no 27

29. The pH value of water is [65th BPSC]

- A. 4 B. 7
C. 12 D. 18
E. None of the above/More than one of the above

Ans. B

Exp: Refer to Que no 27

30. Which of the following has pH value 7? [64th BPSC]

- A. Pure water
- B. Neutral solution
- C. Basic solution
- D. Acidic solution
- E. None of the above/More than one of the above

Ans. E

Exp: Refer to Que no 27

Both the Pure water and Neutral solution can have a pH of value 7. So, more than one of the above answers is correct.

31. A mixture of sand and naphthalene can be separated by [66th BPSC]

- A. Sublimation
- B. Distillation
- C. Chromatography
- D. Fractional distillation
- E. None of the above/More than one of the above

Ans. A

Exp: A mixture of sand and naphthalene can be separated by Sublimation (naphthalene balls sublime).

The term inter-conversion of states is the change of matter from one state to another and back to its original state without a change in its chemical composition.

BCW Bits:

- ❖ **Sublimation:** A change of state directly from a solid to a gas without changing into a liquid state is called sublimation.
- ❖ **Distillation:** It is used for the separation of components of a mixture containing two miscible liquids that boil without decomposition and have sufficient differences in their boiling points.
- ❖ **Chromatography:** It is the technique used for the separation of those solutes that dissolve in the same solvent. It is used to

separate colours in a dye, pigments from natural colours and drugs from the blood.

- ❖ **Fraction distillation:** To separate a mixture of two or more miscible liquids for which the difference in boiling points is less than 25 K, the fractional distillation process is used, for example, for the separation of different gases from the air, different fractions from petroleum products, etc.
- ❖ **Evaporation** is the change of a liquid into vapours at any temperature below its boiling point.
- ❖ **Condensation** is the process through which vapour changes into liquid when it comes in contact with a cooler surface.
- ❖ **Expansion:** An increase in the volume of the substance while its mass remains the same. The anomalous **expansion** of water is an abnormal property of **water** whereby it expands instead of contracting when the temperature decreases from 4° C until it freezes at 0° C. This allows many aquatic life forms to survive through the winter.
- ❖ **Boiling** is the temperature at which the vapour pressure of liquids becomes equal to its atmospheric pressure.
- ❖ **Effervescence:** The formation of gas bubbles in liquid during a chemical reaction is called Effervescence. Bubbles occur as the gas escapes from the reaction.

32. In Interconversion of states of matter: [District Art and Culture BPSC]

- A. Chemical composition changes
- B. Chemical composition does not change
- C. Physical and chemical composition both changes
- D. Physical composition does not change
- E. None of the above/more than one of the above.

Ans. B

Exp: Refer to Que no 31.

33. The conversion of a solid directly into gas is called as [63rd BPSC]

- A. Sublimation
- B. Condensation

- C. Evaporation D. Boiling
E. None of these/More than one of these

Ans. A

Exp: Refer to Que no 31

34. What is condensation? [CDPO 2018 BPSC]

- A. The change of gas into solid
B. The change of solid into a liquid
C. The change of vapour into liquid
D. The change of heat energy into cooling energy
E. None of the above/More than one of the above

Ans. C

Exp: Refer to Que no 31

35. The process of change of state from gaseous to liquid or solid state is called [31st Bihar Judicial Services]

- A. Evaporation B. Condensation
C. Dew Formation D. Liquidization

Ans. B

Exp: Refer to Que no 31

36. The density of water is highest at the temperature of [30th Bihar Judicial Services]

- A. 0° C B. 4° C
C. 50° C D. 100° C

Ans. B

Exp: The density of water is highest at the temperature of 4° C.

BCW Bits:

Thermal expansion of water: It has been observed that Water contracts on heating between 0°C and 4°C temperature. The volume of a given amount of water decreases as it is cooled from room temperature (i.e., when the temperature is decreased from room temperature) until it reaches 4°C.

Below 4°C, the volume increases, and therefore the density decreases (as we know that the Density of a body is mass per unit volume [$d=m/v$], thus the density of a body is inversely proportional to the volume of that body. If the volume of the body increases, the density would decrease and

vice versa).

This property of water is also known as anomalous expansion of water.

Hence, we can say that water has a maximum density at 4°C.

- ❖ This property of the water helps in preserving aquatic life during winter. Water bodies like lakes and ponds start freezing during winter, and they freeze at the top first. As the temperature drops down towards 4°C, water near the surface loses energy to the atmosphere, becomes denser (as explained above) and moves towards the lower layer of the water body and the warmer and less dense water at the bottom layer rises towards the top layer of the water body.
- ❖ When the colder water on top of the water body reaches a temperature below 4°C, water becomes less dense and remains at the surface, where it freezes. In the absence of this property of water, the water bodies would have frozen from bottom to top and aquatic life would have been destroyed during the winters.

37. During very Cold Weather, _____ of water helps in preserving aquatic life [CDPO 2022 BPSC]

- A. Contraction B. Expansion
C. Evaporation D. Sublimation
E. None of the above/More than one of the above

Ans. B

Exp: Refer to Que no 31

38. The process of separating different dissolved constituents of a mixture by their adsorption on an appropriate material is called [CDPO 2022, District Art and Culture]

- A. Diffusion
B. Distillation
C. Chromatography
D. Centrifugation
E. None of the above/More than one of the above

Ans. C

Exp: Refer to Que no 31

In **adsorption**, gasses/liquids to be separated, stick to the surface of the solid.

39. The formation of gas bubbles in liquid during chemical reaction is called [District Art and Culture BPSC]

- A. Evaporation B. Diffusion
C. Condensation D. Effervescence
E. None of the above/more than one of the above.

Ans. D

Exp: Refer to Que no 31

40. A form of matter has no fixed shape but it has a fixed volume. An example of this form of matter is [68th BPSC Pre]

- A. kerosene
B. Carbon steel
C. krypton
D. More than one of the above
E. None of the above

Ans. A

Exp: A form of matter which has no fixed shape but it has a fixed volume is a liquid. Among the given options kerosene is a liquid, hence option A is correct.

BCW Bits:

- ❖ Oxygen is absent in kerosene. It is also called paraffin oil.
- ❖ Krypton (Kr) is a colourless, odourless and tasteless noble gas (Group-18, chemically inert gas) at room temperature with atomic number 36.

41. You have been provided with three test tubes. One of them contains distilled water and the other two contain an acidic solution and a basic solution, respectively. Which of them will turn red litmus to blue? [68th BPSC Pre]

- A. Base
B. Distilled water
C. Acid
D. More than one of the above
E. None of the above

Ans. A

Exp: Base will turn red litmus to blue.

BCW Bits:

Acids → pH value less than 7, higher concentration of H^+ ions. The lower the value of pH stronger the acid.

Base → pH value more than 7, higher concentration of OH^- ions. The higher the value of pH stronger the Base.

Indicators → Special types of substances which are used to test whether a substance is acidic or basic are called Indicators. The indicators change their colour when added to a solution containing an acidic or basic substance.

- ❖ **Litmus** → is the most commonly used natural indicator which is obtained from Lichens.
- ❖ When it is dissolved in an acidic solution it turns red and when added to a basic solution it turns blue. The litmus has purple (mauve) colour in distilled water. It is available in the form of a solution, strips of paper. Hence, only a basic solution can turn red litmus into blue colour.
- ❖ **China Rose** → It is also used as an indicator. It turns acidic solutions to dark pink or magenta and basic solutions to green.
- ❖ **Phenolphthalein** → gives a pink colour when the solution is basic. And when the solution is acidic, it remains colourless.

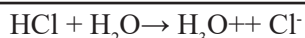
Note: An acid or base reacts with each other, to produce salt (which can be acidic, basic or neutral) and water in the process and the evolution of heat takes place, this reaction is called a **neutralization reaction**.

42. Why does dry HCl gas not change the colour of dry litmus paper? [68th BPSC Pre]

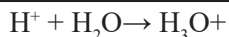
- A. No H_3O^+ ions are present. Litmus changes colour only in the presence of H_3O^+ ions.
B. HCl gas acts as a dehydrating agent
C. Blue litmus becomes dry in the presence of dry HCl gas
D. More than one of the above
E. None of the above

Ans. A

Exp: It has been experimentally observed that hydrogen ion in HCl is produced in the presence of water only. The separation of H^+ ions from HCl molecules cannot occur in the absence of water.



Hydrogen ions cannot exist alone; they exist only after combining with water molecules.



Dry HCl gas acts like dehydrating agent i.e., it removes water which is crucial for forming Hydrogen $[H^+]$ or Hydronium $[H_3O^+]$ ions.

Hence, option A is correct.

43. The acid used in a car battery is [63rd BPSC]

- A. Acetic acid
- B. Hydrochloric acid
- C. Nitric acid
- D. Sulfuric acid
- E. None of these/More than one of these

Ans. D

Exp: Sulphuric acid also called battery acid used in cars consists of 30% - 50% of sulphuric acid in water. The mixture of sulphuric acid and water is sometimes used as an electrolyte in lead-acid batteries.

The old name of sulphuric acid is - **oil of vitriol / the spirit of vitriol / matting acid.**

- ❖ Sulphuric acid is also called the “**king of acids / king of chemicals**” because it is highly corrosive and used in many industries such as explosives, fertilizers, dyes, petroleum refining and metallurgical processes.
- ❖ Sulphuric acid is a component of acid rain formed by atmospheric oxidation of sulphur dioxide in the presence of water.

BCW Bits:

Acid Rain:

Acid rain is caused by a chemical reaction when compounds like sulphur dioxide (SO_2) and nitrogen oxides (NO_x) are released into the atmosphere. These gases rise very high in the atmosphere and react with oxygen and water to form acidic pollutants called acid rain. These

gases are easily soluble in water and form sulphuric acid and Nitric acid. The burning of fossil fuels such as coal in power plants and various vehicular pollutants is responsible for the release of a large amount of sulphur dioxides and nitrogen oxides.

Acid Rain is harmful to crops, rivers, trees, and ancient monuments and structures.

- ❖ Normal rain has a pH value of between 5.0 and 5.5, which is slightly acidic. However, acid rain generally has a pH of around 4. Acid rain is also called **Marble Cancer**.

Note: The term acid rain was coined by **Robert Angus Smith** in 1852.

Q44. The component(s) of acid rain is/are [66 Re BPSC]

- A. HNO_3
- B. H_2SO_4
- C. CO_2
- D. Both (A) and (B)
- E. None of the above/More than one of the above

Ans. D

Exp: Refer to Q no 43

45. Which of the following is present in the maximum amount in acid rain? [62nd BPSC]

- A. HCl
- B. H_2NO_3
- C. H_2SO_4
- D. H_2CO_3
- E. None of the above/More than one of the above

Ans. C

Exp: Refer to Q no 43

46. Consider the following: [UPSC 2022]

1. Carbon monoxide
2. Nitrogen oxide
3. Ozone
4. Sulphur dioxide

Excess of which of the above in the environment is/are causes of acid rain?

- A. 1, 2 And 3
- B. 2 And 4 Only
- C. 4 Only
- D. 1, 3 And 4

Ans. B

Exp: Refer to Q no 43

47. Acid rain is caused by the pollution of the environment by [UPSC 2013]

- A. Carbon Dioxide and Nitrogen
- B. Carbon Monoxide and Carbon Dioxide
- C. Ozone and Carbon Dioxide
- D. Nitrous Oxide and Sulphur Dioxide

Ans. D.

Exp: Refer to Q no 43

48. Which of the following acids does not contain oxygen? [29th Bihar Judicial Service]

- A. Nitric acid
- B. Sulphuric acid
- C. Hydrochloric acid
- D. All of the above

Ans. C

Exp: Hydrochloric acid does not contain oxygen.

The chemical formula of:

- ❖ Nitric acid $\rightarrow \text{HNO}_3$
- ❖ Sulphuric acid $\rightarrow \text{H}_2\text{SO}_4$
- ❖ Hydrochloric acid $\rightarrow \text{HCl}$

Hence, it is clear that hydrochloric acid does not have any oxygen atoms.

49. The chemical name for vinegar is [63rd BPSC]

- A. Methanol
- B. Ethanol
- C. Acetic acid
- D. Ethyl acetate
- E. None of these/More than one of these

Ans. C

Exp: Vinegar is a dilute acetic or ethanoic acid solution, having chemical formula as (CH_3COOH). Acetic acid has the functional group $-\text{COOH}$ (carboxyl group). Production of vinegar depends on the fermentation of ethanol or sugars; it usually contains 4% to 8% of acetic acid and 93-97% of water.

- ❖ Methanol or Methyl alcohol or carbinol (CH_3OH) is a polar solvent, linked to a hydroxyl group ($-\text{OH}$).
- ❖ Methanol is also called **wood spirit** as it is primarily produced by the destructive distillation of wood. It is a colourless liquid soluble in water and forms explosive mixtures with air and burns with non-luminous flame.

Methanol is a toxic alcohol excess intake of methyl causes metabolic acidosis, blindness and death.

Methyl alcohol is used for -

- ❖ Used in radiators of automobiles as antifreeze.
- ❖ Methanol is considered an alternative fuel, as the blending of methyl alcohol with petrol is used as a fuel.
- ❖ Used in the preparation of perfumes, medicines, and dyes and also used as a solvent in the paints, varnish industry, etc.
- ❖ Used in preparation for methylated spirit or denatured spirit (mixture of methyl alcohol & ethyl alcohol) can be used to disinfectant, sterilize medical equipment, etc.

Ethanol-

- ❖ Ethanol is also called ethyl alcohol or alcohol with a hydroxyl ($-\text{OH}$) functional group. Its molecular formula is $\text{C}_2\text{H}_5\text{OH}$. It is produced through fermentation of agricultural products such as sugarcane, corn, wheat etc. by yeast, so it is also called grain alcohol.
- ❖ **Zymase** catalyses the fermentation of glucose to form alcohol (ethanol) and carbon dioxide.
- ❖ Ethanol is a good fuel, a mixture of petrol with ethanol used in motor fuel called power alcohol.
- ❖ It has a high-octane number, making it an anti-knock additive in petrol.
- ❖ It is also used in the chemical industry for the preparation of chloroform, digestive syrups, cough syrup etc.

50. Which of the following is the composition of vinegar? [Headmaster BPSC]

- A. 65% - 75% solution of acetic acid in water
- B. 45% - 60% solution of acetic acid in water
- C. 28% - 35% solution of acetic acid in water
- D. 5% - 8% solution of acetic acid in water
- E. None of the above/More than one of the above

Ans. D

Exp: Refer to Q no 49

51. The odour of acetic acid resembles that of [68th BPSC Pre]

- A. kerosene
- B. vinegar
- C. tomato
- D. More than one of the above
- E. None of the above

Ans. B

Exp: Refer to Q no 49

52. Which acid is described as HOOC-COOH? [68th BPSC Pre]

- A. Acetic acid
- B. Carbonic acid
- C. Oxalic acid
- D. More than one of the above
- E. None of the above

Ans. C

Exp: Oxalic acid is described as HOOC-COOH (or $(\text{COOH})_2$ or $\text{C}_2\text{H}_2\text{O}_4$). It is the simplest dicarboxylic acid. It is an organic compound with the systematic name Ethanedioic acid. It is a weak acid and partially ionized in an aqueous solution. It is widely used for removing stains, marks or rust. It is toxic.

53. Colloids have a particle size range between [CDPO 2022 BPSC]

- A. 10^{-7} m to 10^{-10} m
- B. 10^7 m to 10^{10} m
- C. 10^{-10} m to 10^{-12} m
- D. 10^{-5} m to 10^{-7} m
- E. None of the above/More than one of the above

Ans. A

Exp: Colloids are heterogeneous mixtures in which the particle size is ($10^{-7}\text{m} - 10^{-10}\text{m}$) too small to be seen with the naked eye, but is big enough to scatter light passing through it and make its path visible.

BCW Bits:

- ❖ Colloidal particles can be separated by a technique of centrifugation.

- ❖ Emulsions are liquid colloidal systems in which both the dispersion medium and the dispersed are liquids. It can be separated into their constituent liquids by heating, freezing, centrifuging, etc.

There are two types of emulsions -

- ❖ Oil dispersed in water. In this water acts as a dispersion medium e.g. Milk, vanishing cream, and cod-liver oil.
- ❖ Water dispersed in oil. This oil act as a dispersion medium e.g. Butter, cream
- ❖ Example - Blood, Butter, aerosols, gels, toothpaste

54. Which of the following techniques can be used to separate cream from milk? [Sanitary Waste Management BPSC]

- A. Filtration
- B. Distillation
- C. Centrifugation
- D. Chromatography

Ans. C

Exp: Centrifugation can be used to separate cream from milk. Sometimes the solid particles in a liquid are very small and pass through a filter paper. For such particles, the filtration technique cannot be used for separation. Such mixtures are separated by centrifugation. The principle is that the denser particles are forced to the bottom and the lighter particles stay at the top when spun rapidly.

55. Which among the following is not an example of emulsion? [Headmaster BPSC]

- A. Butter
- B. Chocolate milk
- C. Curd
- D. Whipped cream
- E. None of the above/More than one of the above

Ans. C

Exp: Refer to Q no 54

56. Which of the following is not a characteristic of a chemical change? [CDPO 2022 BPSC]

- A. It is irreversible
- B. New substance is formed
- C. No net change in energy is involved
- D. It involves the absorption of energy

- E. None of the above/More than one of the above

Ans. C

Exp:

Physical change

- ❖ A change in which a substance changes its physical properties is called a physical change.
- ❖ Physical change affects properties such as shape, size, colour, and state of a substance is called physical properties.
- ❖ Physical change is generally reversible, temporary change where no new substance is formed.
- ❖ E.g. Boiling & freezing of water, melting of wax

Chemical Change

- ❖ A change in which one or more new substances are formed is called a chemical change. The chemical change is also called a chemical reaction.
- ❖ Both physical and chemical properties of the substance change.
- ❖ Chemical change is always irreversible, permanent change.
- ❖ New substances are formed
- ❖ Absorption and evolution of energy take place.
- ❖ Involve the production of energy
- ❖ E.g. rusting, fermentation of grapes, digestion of food, ripening

57. Which of the following is chemical change [Headmaster BPSC]

- A. Cooking of foods
- B. Digestion of food
- C. Freezing of water
- D. Water is heated up
- E. None of the above/More than one of the above

Ans. E

Exp: Refer to Q no 56

58. Which of the following are chemical changes?

- (i) Cooking of food
- (ii) Digestion of food
- (iii) Freezing of water
- (iv) water is heated up

- A. i and ii
- B. i, ii and iii
- C. i and iv
- D. ii, iii and iv

Ans. A

Exp: When a substance undergoes a physical change, its composition remains the same despite its molecules being rearranged. Physical change is a temporary change. Whereas, when a substance undergoes a chemical change, its molecular composition is changed entirely. Thus, chemical changes involve the formation of new substances. A chemical change is a permanent change.

For more details refer to Que no 56

59. Which of the following is/are the example/examples of chemical change? [UPSC 2014]

- 1. Crystallization of sodium chloride
- 2. Melting of ice
- 3. Souring of milk

Select the correct answer using the code given below.

- A. 1 And 2 Only
- B. 3 Only
- C. 1, 2 And 3
- D. None

Ans. B

Exp:

- ❖ Crystallization of NaCl is a physical process as in crystallization no new product is formed; only the salt is obtained in its crystal form by allowing the evaporation of the Solvent.
- ❖ Similarly, in the process of melting ice, only the form of water is changing from solid to liquid. So, it is a physical change.
- ❖ In the souring of milk, the formation of lactic acid takes place by the action of the bacteria present in the milk. A new substance is formed and we cannot obtain milk from the acidified milk hence, it is a chemical change.

60. In endothermic reactions [Bihar Assistant Prosecution Officer]

- A. Heat Is Absorbed
- B. Heat Is Evolved

- C. Temperature Is Increased
D. Light Is Produced

Ans. A

Exp: There are various types of chemical reactions such as combination reactions, decomposition reactions, exothermic reactions, endothermic reactions, displacement reactions, double-displacement reactions, etc.

Endothermic reaction → Reaction in which energy is absorbed is called endothermic reaction.

Eg.: Decomposition of calcium carbonate to calcium oxide and carbon dioxide, decomposition of silver chloride into silver and chlorine by light (it is used in black and white photography).

Hence, option A is correct.

BCW Bits:

- ❖ Combination Reaction: → Two or more substances combine to form a new single substance.
- ❖ Eg– burning of coal, formation of water from hydrogen and oxygen atom, etc
- ❖ Decomposition Reaction: → These are opposite to combination reactions. In a decomposition reaction, a single substance decomposes to give two or more substances.
- ❖ Exothermic Reaction: → Reaction in which energy is given out with the formation of a product is called exothermic reaction. Eg– reaction of calcium oxide (slaked-lime) with water, Burning of Natural Gas, respiration, Decomposition of vegetable matter into compost, etc.
- ❖ Displacement Reaction: → When an element displaces another element from its compound, the reaction involved is called displacement reaction.
- ❖ Double-Displacement Reaction: → When two or more atoms or ions get displaced, the reaction involved is called Double-Displacement Reaction.
- ❖ Oxidation → gain of an oxygen atom or loss of a hydrogen atom.
- ❖ Reduction → loss of an oxygen atom or gain of a hydrogen atom.

61. Detect the odd one [68th BPSC Pre]

- A. Burning of coal
B. Ripening of fruit
C. Setting of cement
D. More than one of the above
E. None of the above

Ans. E

Exp: Burning of coal, ripening of fruit, setting of cement → all of them are chemical changes and all are also exothermic. Hence, the correct option is 'E'.

BCW Bits:

- ❖ Chemical change: It can be defined as a change in which one or more new substances are formed. Eg: digestion of food in our body, ripening of fruits, fermentation of grapes, etc.
- ❖ Exothermic Reaction: Reaction in which energy is given out with the formation of a product is called exothermic reaction. Eg– reaction of calcium oxide (slaked-lime) with water, Burning of Natural Gas, respiration, Decomposition of vegetable matter into compost, etc.
- ❖ Burning of coal: When the burning of coal takes place in the presence of air it produces carbon dioxide, heat and light.
- ❖ Ripening of fruit: In the process of ripening of fruit, the rate of respiration increases and heat is released in the respiration process, hence, ripening of fruit is an exothermic process. Ethylene is highly effective in fruit ripening. It enhances the respiration rate during the ripening of fruits. The process of rise in the rate of respiration is called respiratory climactic.
- ❖ The setting of cement:
 - Cement: it is an important building material. It is obtained from a combination of a material having high lime content, CaO and other materials such as Clay which contains silica, SiO₂ and oxides of aluminum, iron and magnesium.
 - It was first introduced by **Joseph Aspdin** in 1824 in England.

- **Formation:** When clay and lime are strongly heated together, they fuse and react together they fuse and react to form a 'cement clinker'. This clinker is mixed with 2-3% (by weight) of gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) to form cement.
- **Main components (Portland cement):** dicalcium silicate (Ca_2SiO_4)-26%, tricalcium silicate (Ca_3SiO_5) - 51% and tricalcium Aluminate ($\text{Ca}_3\text{Al}_2\text{O}_6$)-11%.

❖ When water is added to the cement, the setting of cement takes place to give a hard mass. This happens due to the hydration of the molecules of the constituents and their rearrangement.

62. Which of the following is not a compound? [Sanitary Waste Management BPSC]

- A. Ammonia B. Milk
C. Sodium chloride D. Chalk

Ans. B

Exp: Milk is a mixture as it consists of water and fats which are present together but they are not chemically combined. The composition also may or may not be fixed that's why it is a mixture.

❖ Compound is a pure substance which consists of two or more than two atoms combined chemically in a fixed ratio by mass

Test Your Knowledge

1. Which one of the following processes involves chemical reactions?

- A. Storing of oxygen gas under pressure in a gas cylinder
B. Liquefaction of air
C. Heating copper wire in the presence of air at high temperatures
D. More than one of the above
E. None of the above

Ans. C

2. Consider the following statement regarding Acetic Acid

- I. The chemical formula is $\text{C}_2\text{H}_4\text{O}_2$
II. Also known as king of all Acids

III. It is used in petroleum refining
Which among the above is/are correct

- A. I only
B. II and III only
C. I and III only
D. More than one of the above
E. None of the above

Ans. A

3. Consider the following statements:

- (i) An acid turns blue litmus to red
(ii) Methyl orange gives red colour in an acid solution
(iii) Phenolphthalein remains colourless in an acid
(iv) Phenolphthalein gives yellow colour in a basic solution.

Which of the following statements is/are correct?

- A. (i) only
B. (i) and (ii) only
C. (i), (ii) and (iii) only
D. More than one of the above
E. None of the above

Ans. C

4. Consider the following statements:

1. A change of state directly from solid to gas without changing into a liquid state is called sublimation.
2. The direct change of gas to solid without changing into liquid is called deposition.
3. The phenomenon of change of a liquid into vapours at any temperature above its boiling point is called evaporation.
4. The change of a solid state into a liquid state is known as fusion.

Which of the statements given above is/are correct?

- A. 1 and 2 only
B. 1, 2, and 3 only
C. 1, 2, and 4 only
D. More than one of the above
E. None of the above

Ans. C

METALS, MINERALS, ORES, ALLOYS

63. The metallurgical process in which metal is obtained in a fused state is called [67Re BPSC]

- A. Roasting B. Calcination
C. Froth flotation D. Smelting
E. None of the above /more than one of the above

Ans. D

Exp: **Smelting** is the process in which the oxide of metal is heated with a reducing agent and flux in a blast furnace. Therefore, metal is obtained in a fused state.

- ❖ **Roasting:** In roasting, the ore is heated in a regular supply of air in a furnace at a temperature below the melting point of the metal.
- ❖ **Calcination:** Calcination involves heating when the volatile matter escapes leaving behind the metal oxide
- ❖ **Froth flotation:** It is a method for removing gangue from sulphide ores. In this process, a suspension of the powdered ore is made with water. To it, collectors and froth stabilisers are added. The invention of the Froth Floatation Method made copper mining profitable even from the low-grade ores.

64. Which among the following are constituents of brass? [67Re BPSC]

- A. iron and zinc B. copper and nickel
C. iron and copper D. zinc and copper
E. none of the above more than one of the above

Ans. D.

Exp: An alloy is a homogeneous mixture of two or more metals, or a metal and a Non-metal; It is prepared by melting primary metal. Alloying metal is a method of improving the properties of metal and properties can also be changed.

Alloy	Composition	Uses
Brass	Copper + Zinc	Locks, gears, bearings, zippers, electrical plugs and sockets and valves, etc.

Alloy	Composition	Uses
Bronze	Copper + Tin	Coins, utensils, automobile parts, etc.
Steel	Iron + Carbon	Bridges, buildings, infrastructure, machines, etc.
Stainless steel	Steel + Nickel + Chromium + manganese	Utensils, cutlery, surgical equipment in architecture, aerospace applications, storage tanks etc.
Solder	Tin + Lead	Air conditioning, fire sprinkler, electronics, etc.
German silver	Copper + Zinc + Nickel	Decorative articles, ornaments, musical instruments, etc.
Gun metal	Copper + Zinc + Tin	Machine parts, guns, valves, gear, etc.
Fuse wire	Tin + Lead	Fuse
Duralumin	Aluminum + Manganese + copper + Magnesium Iron	Airplanes, spacecraft, ships etc.
Nichrome	Nickel + chromium + Iron	Resistance wire, Iron and heater coils, electric kettle, etc.

65. Brass is an alloy of [CDPO 2018 BPSC]

- A. Nickel and copper
B. Copper and silver
C. Nickel and zinc
D. Zinc and copper
E. None of the above/More than one of the above

Ans. D

Exp: Refer to Que no 64

66. Stainless steel is an alloy of [63rd BPSC]

- A. Iron and nickel
B. Iron and chromium
C. Copper and chromium
D. Iron and zinc

- E. None of the above/More than one of the above

Ans. E

Exp: Refer to Que no 64

BCW Bits:

- ❖ Chromium protects steel from corrosion and provides high-temperature resistance.
- ❖ By adding nickel to alloy, stainless steel becomes non-magnetic.
- ❖ Carbon is mixed to increase hardness and strength, an excess percentage of carbon increases brittleness and reduces weldability.
- ❖ Manganese added less than 1% is used as a substitute for nickel.

67. Stainless steel is an alloy of iron mainly with [Bihar Assistant Prosecution Officer]

- A. Zinc B. Copper
C. Chromium D. Magnesium

Ans. C

Exp: Refer to Que no 64

68. Which of the following is incorrect? [Headmaster BPSC]

- A. Alloying helps in improving the properties of a metal.
B. An alloy is a heterogeneous mixture of two or more metals.
C. The melting point of an alloy is less than that of a pure metal.
D. Some familiar examples of alloy include brass, bronze and steel.
E. None of the above/More than one of the above

Ans. B

Exp: Refer to Que no 64

69. What happens to the weight of Iron, when it rusts? [67Re BPSC]

- A. Decreases then increases
B. Increases then decreases
C. Remain same
D. Increase for a long time

- E. None of the above /more than one of the above

Ans. D

Exp: Corrosion results in the conversion of the metal into its chemically stable form which can either be oxides or hydroxides or sulphides.

Rusted Iron weight = Weight of Iron + Weight of oxygen

Due to the addition of oxygen during rusting, we need to take into account the weight of oxygen now attached to the Iron, hence the weight of rusted iron increases.

Rusting is a type of chemical change. It results in the formation of iron oxides which is an entirely new substance. It is an example of oxidation reaction.



BCW Bits:

Prevention from rusting —

- ❖ **Alloying** – alloying (mixing) the iron with other stable metals or alloys can slow down the process of rusting.
- ❖ **Galvanizing** – Applying a protective layer of metallic zinc on steel or iron to prevent rusting is called **Galvanization**, it is an inexpensive procedure. **Zinc is used** for galvanizing because (creates a sacrificial anode) zinc is more reactive to oxygen than iron, so it oxidizes more quickly and protects from rusting.
- ❖ **Coating and painting** – coating the surface of a metal object with a layer of either paint or varnish will work as a barrier between the surface and atmospheric oxygen making it immune to rusting.

70. The rusting of iron is an example of [Bihar Assistant Prosecution Officer]

- A. Oxidation B. Reduction
C. Polymerization D. Galvanization

Ans. A

Exp: Refer to Que no 69

71. Galvanized iron pipes have a coating of [63rd BPSC]

- A. Zinc B. Mercury

- C. Lead D. Chromium
E. None of these/More than one of these

Ans. A

Exp: Refer to Q no 69

72. The purest form of iron is [CDPO 2018 BPSC]

- A. Steel B. Pig iron
C. Cast iron D. Wrought iron
E. None of the above/More than one of the above

Ans. D

Exp: Wrought iron or malleable iron is the purest form of commercial iron with less carbon content (less than 0.08% of carbon) and is prepared from cast iron by oxidizing impurities in a reverberatory furnace lined with haematite.

It is soft, ductile magnetic and has high elasticity and tensile strength. It is used for making wires, chains, and agricultural implements.

Cast iron is made by melting pig iron with scrap iron and coke using a hot air blast. It has a slightly lower carbon content about 3% and is extremely hard and brittle. It is used for casting stoves, railway sleepers, toys etc.

Metal	Ore	Composition
Iron	❖ Haematite	Fe_2O_3
	❖ Magnetite	Fe_3O_4
	❖ Siderite	FeCO_3
	❖ Iron pyrites	FeS_2

73. Which among the following is also known as white metal? [67Re BPSC]

- A. Rhodium B. Platinum
C. Palladium D. Nickel
E. None of the above/ more than one of the above

Ans. E

Exp: Palladium is a chemical element with the symbol Pd and atomic number 46. It is a rare and lustrous silvery-white metal discovered in 1803 by **William Hyde Wollaston**.

Platinum is naturally white metal and due to its white appearance, it is called "White Gold".

Hence the correct option is 'E'.

74. Consider the following statements with respect to noble metals [67Re BPSC]

- Noble metals are found in pure form in nature.
- Uranium and lead are examples of noble metals.

Which of the above statements is correct?

- A. 2 only
B. Both 1 and 2
C. Brass is also noble metal
D. 1 only
E. None of the above/ more than one of the above

Ans. D.

Exp: Noble metals are metals that have very low reactivity. They are found in the sixth and seventh columns of the periodic table. Examples of noble metals are gold, silver, platinum, and palladium. These metals are very resistant to corrosion and oxidation. They are used in jewellery, coins, and other decorative items.

75. The heaviest metal among the following is [63rd BPSC]

- A. Gold B. Silver
C. Mercury D. Platinum
E. None of these/More than one of these

Ans. D

Exp: Heavy metals have a relatively high density, atomic weights or atomic numbers. Metals with low density are called light metals.

Lithium is the lightest metal and **Osmium** is the heaviest.

Metal	Atomic symbol	Density
Gold	Au	19.3
Silver	Ag	10.5
Mercury	Hg	13.6
Platinum	Pt	21.45

76. The heavier metal among the following is [Bihar Assistant Prosecution Officer]

- A. Iron B. Mercury
C. Nickel D. Osmium

Ans. D

Exp: Refer to Q no 75

77. Identify the metal which is non-toxic in nature.
[30th Bihar Judicial Services]

- A. Chromium B. Gold
C. Cadmium D. Cobalt

Ans. B

Exp: Gold is non-toxic in nature.

BCW Bits:

Toxic substances are those which have adverse effects on the environment or mankind. Metals like Chromium, Cadmium and Cobalt are toxic.

- ❖ Chromium (Cr): Exposure to this metal can occur from natural or industrial sources. When inhaled it causes respiratory tract irritation. It can cause diseases like asthma, Bronchitis, etc.
- ❖ Cadmium (Cd): If a person is exposed to cadmium present in the air, food, water and particularly in tobacco smoke over time can lead to cadmium buildup in the kidney and cause kidney diseases and fragile bones. **Itai-Itai** disease is caused by cadmium poisoning.
- ❖ Cobalt (Co): It is a hard, grey and naturally occurring metal. It can cause harm to the eyes, skin, heart, and lungs. Prolonged exposure to cobalt may cause Cancer as well.
- ❖ Mercury (Hg): poisoning causes **Minamata disease**.
- ❖ Arsenic: (As) poisoning causes skin-related disease and cancer.

78. The most important ore of aluminum is [67 Cancelled BPSC]

- A. Bauxite B. Calamite
C. Calcite D. Galena
E. None of the above/more than one of the above

Ans. A

Exp:

Metal	Ores
Aluminum	❖ Bauxite
	❖ Cryolite
	❖ Corundum
	❖ Feldspar
Magnesium	❖ Magnesite
	❖ Dolomite
	❖ Carnallite
Zinc	❖ Calamine
	❖ Zincite
	❖ Sphalerite
Mercury	❖ Cinnabar
	❖ Galena
	❖ Cerussite
Lead	❖ Anglesite

79. The chemical formula for Sapphire (Ruby) [66 Re BPSC]

- A. Al_2O_3 B. Al_3O_2
C. N_2O D. NO_2
E. None of the above/ More than one of the above

Ans. A

Exp: Sapphire (Al_2O_3) is a variety of mineral Corundum. The Corundum is naturally found in crystals of aluminum oxide typically containing traces of Iron, Titanium, Vanadium and Chromium. The addition of chromium oxide gives it a **ruby (red)** colour.

BCW Bits:

- ❖ Ruby is defined as red corundum. The presence of chromium is largely what makes a corundum gemstone red. All other varieties of corundum, anything not red, are classified as sapphire. (Sapphires may contain a mix of chromium, titanium, and iron traces).
- ❖ Rubies and Sapphires are the same mineral (corundum). Both are made of alumina and oxygen, but they just differ in color. When a corundum is **Red**, it is classified as a ruby, and when it's **Blue**, it is called a sapphire.

80. Which of the following has the highest melting point? [66 Re BPSC]

- A. Boron B. Iron
C. Silicon D. Aluminium
E. None of the above/More than t of the above

Ans. A

Exp: Boron forms a covalent bond with the neighbouring elements by sharing its valence electrons with the other elements. A covalent bond is a strong bond and requires high energy to break it thereby possessing a higher melting point among the metalloid.

BCW Bits:

Metalloids can be defined as chemical elements which have chemical properties and physical properties that fall in between those of metals and non-metals categories. Group 13 is considered as metalloids.

There are 7 metalloids Boron, Silicon, Germanium, Arsenic, Antimony, Tellurium, Polonium, and Astatine.

Boron has a brownish-red colour. It is in solid form at Room temperature and evaporates easily. Boron has high melting and boiling point compared to other elements in group 13 because of strong covalent bonds between the atoms of boron.

81. Which of the following is not a type of element? [65th BPSC]

- A. Metals B. Non-metals
C. Gases D. Metalloids
E. None of the above/More than one of the above

Ans. C

Exp: Refer to Q.No. 80

Gas is a state of matter. An element can exist in any of the three states of the matter i.e., solid, liquid or gas.

82. Which is the most reactive metal? [59th BPSC]

- A. Sodium B. Calcium
C. Iron D. Potassium

Ans. D

Exp: **Reactivity series/activity series** is a list of metals arranged in order of decreasing reactivity.

- ❖ Most reactive metals at the top of the series are powerful reducing agents. (Reducing agents lose or donate electrons and oxidize in chemical reactions.)
- ❖ All metals that are placed above hydrogen react with dilute hydrochloric acid or sulphuric acid to liberate hydrogen.
- ❖ $\text{Metal} + \text{Acid} \rightarrow \text{salt} + \text{Hydrogen}$
- ❖ Examples of some highly reactive metals are - Francium, potassium, sodium, lithium, calcium
- ❖ Examples of least reactive metals are- copper, silver, gold, platinum

Metals	Reactivity
Potassium	Reacts with water
Sodium	
Lithium	
Barium	
Strontium	
Calcium	
Magnesium	Reacts with acids
Aluminium	
Manganese	
Zinc	
Chromium	
Iron	
Cadmium	
Cobalt	
Nickel	
Tin	
Lead	Included for comparison
Hydrogen	
Antimony	Highly unreactive
Bismuth	
Copper	
Mercury	
Silver	
Gold	
Platinum	

83. Silver, Iron and Copper are reactive in nature. Which of the following is the correct order [District Art and Culture BPSC]

- A. Silver>Copper>Iron
- B. Copper>Iron>Silver
- C. Silver>Iron>Copper
- D. Iron>Silver>Copper
- E. None of the above/more than one of the above.

Ans. E

Exp: Refer to Que no 82

Test Your Knowledge

1. Copper and zinc alloy is known as

- A. Brass
- B. Bronze
- C. Duralumin
- D. More than one of the above
- E. None of the above

Ans. A

2. Consider the following pairs:

Match-I (Homogeneous mixture)	Match-II (Alloy)
1. Mercury and any other metal	Amalgam
2. Copper and Zinc	Brass
3. Copper and tin	Bronze
4. Lead and Tin	Solder

Which of the following pairs given above is/are correctly matched?

- A. 1 and 4 only
- B. 1 and 2 only
- C. 1, 2, 3 and 4
- D. More than one of the above
- E. None of the above

Ans. C

3. Consider the following statements about Iron:

- I. Cast iron is an impure form of iron
 - II. Wrought iron is the pure form of iron
 - III. Cast iron is used in making railway sleepers
 - IV. Wrought iron is not obtained by Cast iron
- Which of the statements given above is/are correct?

- A. I and II Only
- B. I, II and III Only
- C. I and IV Only
- D. More than one of the above
- E. None of the above

Ans. B

4. Magnalium is an alloy of

- A. Aluminium and Magnesium
- B. Magnesium and silver
- C. Iron and copper
- D. More than one of the above
- E. None of the above

Ans. A

NON-METAL COMPOUNDS

84. Which of the following statements about graphite and diamond is true? [67 cancelled BPSC]

- A. They have the same crystal structure.
- B. They have the same hardness.
- C. They have the same electrical conductivity.
- D. They can undergo the same chemical reaction
- E. None of the above/more than one of the above

Ans. D

Exp: Carbon is found in all **organic compounds** and the atomic number of carbons is 6.

❖ **Dry ice** is known as solid CO_2 .

❖ Carbon has three main **allotropes** -

A) **Graphite** – It occurs in the most stable form of carbon, arranged in a hexagonal structure, it is a **good conductor of heat**, soft and has a hardness of 1 to 2 on the Mohs scale. It is used in pencils and as a **Dry Lubricant** in machines at high temperatures.

❖ **Graphite and diamond are identical chemically** (both are composed of carbon), but physically, they are very different (**different crystal structures**-polymorphs).

❖ Graphite is opaque and is a non-metal as it is made up of carbon (although it does show

metallic properties like “lustre”), earthy looking, while diamonds are transparent and brilliant.

B) Diamond – It is arranged in a tetrahedral structure, has a high value of the refractive index and is extremely hard (strong covalent bonds), it is a **bad conductor of electricity**, an excellent insulator and a **good conductor of heat**. Diamonds are used for rock drilling, grinding and cutting metals and glass.

C) Buckminsterfullerene - C_{60} is made up of 20 hexagons and 12 Pentagons, used as lubricant and catalyst, and also used for several biomedical applications such as the design of high-performance MRI contrast agents, and X-ray imaging contrast agents.

❖ Fullerenes (C_{60}) and their derivatives have potential antiviral activity, which has strong implications for the treatment of HIV infection.

Note: The purest form of Carbon is “Fullerene” and it was discovered by **Kroto, Curl, and Smalley** in 1985.

85. Which of the following is used in ordinary pencils? [Bihar Assistant Prosecution Officer]

- A. Graphite B. Silicon
C. Lead D. Charcoal

Ans. A

Exp: Refer to Que no 84

86. Which of the following statements about diamond and graphite is true? [68th BPSC Pre]

- A. They have the same degree of hardness
B. They have the same electrical conductivity
C. They have the same crystal structure
D. More than one of the above
E. None of the above

Ans. E

Exp: Refer to Que no 84

87. Which of the following elements is found in all organic compounds? [59th BPSC]

- A. Carbon B. Calcium
C. Nitrogen D. Oxygen

Ans. A

Exp: Refer to Que no 84

88. The purest form of the amorphous carbon is [CDPO 2022 BPSC]

- A. Wood charcoal B. Sugar charcoal
C. Bone charcoal D. Lampblack
E. None of the above/More than one of the above

Ans. D

Exp: Lampblack contains (98-99) % of carbon in amorphous form. It can be easily obtained by heating carbon-rich substances like kerosene oil in a limited supply of oxygen. So, the correct answer is option ‘D’.

89. The numbers of sigma and pi bonds in benzene are [66th BPSC]

- A. 3, 3 B. 3, 6
C. 12, 3 D. 12, 6
E. None of the above/More than one of the above

Ans. C

Exp: **Structure of Benzene:**

So, the number of sigma bonds is 12 and the number of pi bonds is 3.

❖ **Sigma Bond:** This type of covalent bond is formed by the end-to-end (head-on) overlap of bonding orbitals along the internuclear axis.

❖ **Pi bond:** In the formation of pi (π) bonds the atomic orbitals overlap in such a way that their axes remain parallel to each other and perpendicular to the internuclear axis.

90. What is the bond order of the CO group? [65th BPSC]

- A. 1 B. 2.5
C. 3.5 D. 3
E. None of the above/More than one of the above

Ans. D

Exp: The bond order of CO is 3. The bond order shows the number of chemical bonds present between a pair of atoms.

91. A hydrocarbon in which two carbon atoms are joined by a double bond is called an [63rd BPSC]

- A. alkane B. alkene
C. alkyne D. ionic bond
E. None of these/More than one of the

Ans. B

Exp: Alkene are a class of hydrocarbons containing **double bonds (C=C)** between carbon atoms, also known as olefins and the general molecular formula C_nH_{2n} e.g. - C_2H_4 (ethene), C_3H_6 (propene), C_4H_8 (butene)

BCW Bits:

- ❖ **Alkanes** are saturated hydrocarbons in which carbon atoms are bonded by single **covalent bonds (-C-C-)**, because of the covalent nature of C-C and C-H bonds alkanes are nonpolar molecules and weak **Vander Waal force** exists between bonds. They are also called paraffins because they are less reactive towards other reagents. Molecular formulas for alkanes are - C_nH_{2n+2}
- ❖ e.g. - CH_4 (methane), C_2H_6 (Ethane), C_3H_8 (propane).
- ❖ **Alkynes** are unsaturated hydrocarbons having a triple bond between carbon atoms and **triple bond** known as an acetylenic bond their general formula is C_nH_{2n-2} e.g. C_2H_2 (Ethyne), C_3H_4 (Propyne), C_4H_6 (Butyne).

92. Oxygen is absent in [66Re, 65th BPSC]

- A. Glass B. Soil
C. Cement D. Kerosene
E. None of the above/More than one of the above

Ans. D

Exp: Oxygen is absent in kerosene, called paraffin or paraffin oil, which is a flammable hydrocarbon liquid commonly used as a fuel. Its chemical formula is $C_{12}H_{26}C_{15}H_{32}$.

- ❖ The chemical formula of most glass is SiO_2 .
- ❖ The main ingredients of cement are CaO , SiO_2 , Al_2O_3 , Fe_2O_3 , MgO , etc.
- ❖ Oxygen is found either dissolved in soil solution or in the atmosphere.

So, from the above description, it is clear that kerosene doesn't have oxygen in it.

93. Some metal oxides react with acid as well as base to produce salt and water. They are known as amphoteric oxides. The examples of amphoteric oxides are [68th BPSC Pre]

- A. Al_2O_3 and CuO
B. CuO and ZnO
C. Al_2O_3 and ZnO
D. More than one of the above
E. None of the above

Ans. C

Exp: The examples of amphoteric oxides are, Al_2O_3 and ZnO .

BCW Bits:

Oxides → A chemical compound which consists of two elements in which one of which is oxygen is called an oxide. An oxide varies widely in nature and properties. They can be simple (such as MgO , Al_2O_3 etc) or mixed (Pb_3O_4 , Fe_3O_4).

Simple oxides can be acidic, basic or amphoteric.

Acidic oxides are those oxides which when combined with water give an acid as the product.

Eg → SO_2 ($SO_2 + H_2O \rightarrow H_2SO_3$, which is an acid), Cl_2O_7 , CO_2 , N_2O_5 . In general, non-metallic oxides are acidic but oxides of some metals in high oxidation state do show an acidic character (eg → Mn_2O_7 , CrO_3 , V_2O_5).

Basic oxides are those oxides which when combined with water give a base as a product.

Eg → Na_2O , CaO ($CaO + H_2O \rightarrow Ca(OH)_2$), BaO , etc. Generally, metallic oxides are basic.

Amphoteric oxides → some metal oxides show dual nature, which means they show properties of both acidic as well as basic oxides. They are known as amphoteric oxides. They react with acids as well as bases to produce salt and water. Eg → Al_2O_3 and ZnO .

Neutral oxides → These oxides are neither acidic nor basic.

Eg → CO , NO and N_2O .

Q94. Which of the following is also known as laughing gas? [65th BPSC]

- A. Nitric oxide B. Nitrous oxide
C. Nitrogen pentoxide D. Nitrogen
E. None of the above/More than one of the above

Ans. B

Exp: Nitrous oxide (N_2O), is also known as **Laughing gas**. It is a colourless, odourless, non-flammable gas, it supports combustion more vigorously than air. It is most commonly used anesthesia and also used in sedatives provided by dentist to patients. The body's reaction time slows down when the gas is inhaled, resulting in a calm and euphoric feeling. Laughing gas functions as a mild sedative but can also be used to treat pain.

- ❖ Nitrous oxide was discovered by **Joseph Priestley** in 1772 and later **Humphry Davy** coined the name laughing gas.

BCW BITS:

- ❖ Nitrogen - (N_2) makes up 78.1% of the earth's atmosphere, it is inert gas (requires more energy to react with other substances) therefore aircraft tyres are filled with nitrogen.
- ❖ Nitrogen is essential for growth and reproduction. It is a major component of amino acids, the building blocks of proteins.
- ❖ Insectivorous plants grow in nitrogen-deficient soil and get nitrogen from insects they trap.
- ❖ Nitrogen gas is used in chips packets because it controls the Oxygen reactivity. The agents called antioxidants prevent oxidation of oils and fats in food. Nitrogen is used to flush bags of chips because it slows down the oxidation of chips.

Note: N_2O is both a Greenhouse gas and an ozone-depleting substance.

Q.95. The gas, which comes out on opening a soda water bottle, is [63rd BPSC]

- A. Carbon dioxide B. Hydrogen
C. Nitrogen D. Sulfur dioxide
E. None of the above/More than one of the above

Ans. A

Exp: Carbon dioxide comes out on opening a soda water bottle, as gas is dissolved in soda under

high pressure. So, when we open the bottle, the pressure inside the bottle decreases, therefore, the solubility of CO_2 decreases, and gas fizzes out.

- ❖ Solubility of gas in a liquid is directly proportional to the pressure of the gas.
- ❖ **Henry's Law** states that at a constant temperature, the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas present above the surface of the liquid or solution.

96. Soda water obtained by passing carbon dioxide in water is [29th Bihar Judicial Service]

- A. An oxidizing agent
B. Basic in nature
C. Acidic in nature
D. A reducing agent

Ans. C

Exp: Soda water is produced by passing carbon dioxide into water; as a result, the formation of carbonic acid (a weak acid) takes place.

$[H_2O(l) + CO_2(g) \rightarrow H_2CO_3(aq)]$. Therefore, soda water is acidic.

Q97. Gases used in welding are [63rd BPSC]

- A. Oxygen and hydrogen
B. Oxygen and nitrogen
C. Oxygen and acetylene
D. Hydrogen and acetylene
E. None of the above/More than one of the above

Ans. C

Exp: Oxygen and acetylene are used in welding. Acetylene is the simplest alkyne (C_2H_2). It is widely used as fuel and when it reacts/burns with oxygen it produces enough energy to melt steel (nearly 1500 degree Celsius) hence it is used in welding.

98. Bromine is a [63rd BPSC]

- A. Colorless gas
B. Brown solid
C. Highly inflammable gas

- D. Red liquid
E. None of the above/More than one of the above

Ans. D

Exp: Bromine is a reddish-brown liquid, part of the halogen group. It is used as sedatives, agricultural chemicals, insecticides, dyes, fire retardants, gasoline, and plastic casings for electronics, and silver bromide is used in photography.

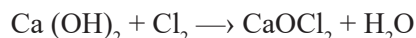
Bromine is the only non-metal to exist as a liquid at room temperature.

99. What is the chemical name of bleaching powder? [Headmaster BPSC]

- A. Calcium oxychloride
B. Calcium monosulphate
C. Calcium tetrachloride
D. Sodium chloride
E. None of the above/More than one of the above

Ans. A

Exp: Chemical formula of bleaching powder is CaOCl_2 . It is prepared when chlorine gas is placed over slaked lime (Ca(OH)_2).



Uses —

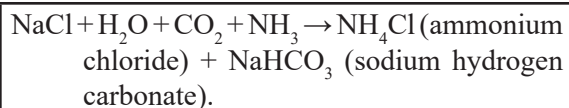
- ❖ In the textile industry for bleaching cotton and linen
- ❖ In paper factories for bleaching wood pulp
- ❖ In laundry for bleaching washed clothes
- ❖ For disinfecting drinking water to make it free of germs.
- ❖ As an oxidizing agent in many chemical industries.

100. Baking Soda is [30th Bihar Judicial Services]

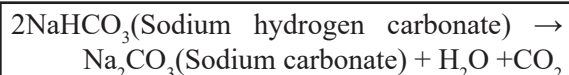
- A. Sodium carbohydrate
B. Sodium bicarbonate
C. Sodium sulfate
D. Sodium hydroxide

Ans. B

Exp: The chemical name of baking soda is **Sodium Hydrogen Carbonate (NaHCO_3)** also known as **sodium bicarbonate**.

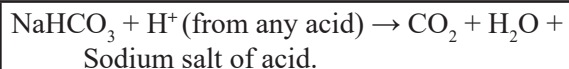


It is a mild non-corrosive basic salt; following reaction takes place when it is heated during cooking:



Uses: →

- ❖ It is commonly used in the kitchen for making food crispy.
- ❖ It is used for making baking powder, which is a mixture of Sodium Hydrogen Carbonate (Baking Soda) and a mild edible acid such as tartaric acid.
- ❖ When Baking powder is heated or mixed in water, carbon dioxide is produced which causes bread or cake to rise and makes them soft and spongy.



- ❖ It also acts as an antacid. Being alkaline in nature, it neutralizes excess acid in the stomach and provides relief.
- ❖ It is also used in soda-acid fire extinguishers.

101. Which one of the following has the highest melting point? [30th Bihar Judicial Services]

- A. Gold B. Silver
C. Silicon D. Copper

Ans. C

Exp: Silicon is a non-metal having a giant covalent structure. It is a p-block, Group-14 element. Elements of this group have strong 'Catenation' properties. And due to high electronegativity, silicon has high bond enthalpy ($-\text{Si-Si}- \rightarrow 297 \text{ kJ/mol}$).

- ❖ Catenation → The self-linking property of an element, to form long chains or branched chains of the atoms of that element or they could be arranged in rings. Carbon exhibits the highest catenation tendency.

- The order of catenation is $\rightarrow \text{C} > \text{Si} > \text{Ge} = \text{Sn}$.

- ❖ Electronegativity → is the tendency to attract shared pairs of electrons.

Thus, atoms of silicon have strong covalent bonds between them and a large amount of energy is required to break that bond. So, silicon has the highest melting point among the given options. All other elements in the option are metals.

102. Quartz crystals used in clocks are [63rd BPSC]

- A. Sodium silicate B. Silicon dioxide
C. Germanium oxide D. Titanium dioxide
E. None of these/More than one of these

Ans. B

Exp: Quartz crystals in clocks are silicon dioxide and the chemical formula is SiO_2 .

Note: 1. Most abundant element present in the earth's crust is oxygen (46%) followed by silicon (28.2%)
2. Silicon was discovered by Berzelius.

103. Which of the following is an amorphous solid? [Sanitary Waste Management BPSC]

- A. Quartz Glass (SiO_2)
B. Chrome Alum
C. Silicon Carbide (SiC)
D. Graphite

Ans. A

Exp: If quartz is melted and the melt is cooled rapidly enough to avoid crystallization an amorphous solid called glass is obtained.

The solids in which the particles are not arranged in any specific order or the solids that lack the overall order of a crystal lattice are called **amorphous solids**.

104. Blue vitriol is chemically [63rd BPSC]

- A. Sodium sulfate
B. Nickel sulfate
C. Copper sulfate
D. Iron sulfate
E. None of these/More than one of these

Ans. C

Exp: Blue vitriol (Neela thotha) is a compound of copper sulphate whose chemical formula is

$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.

Copper sulphate is used as a fungicide, herbicide, and algacide. Copper sulphate is also used for water purification or sterilization systems because of its antibacterial effect.

Note: The number of water molecules attached to one formula unit is called water of crystallization. So, in the case of blue vitriol, we have '5' water molecules attached to one formula unit, hence water of crystallization will be 5.

105. The chemical used for destroying fungi in the water tank is [66 Re BPSC]

- A. Nitric acid
B. Zinc sulphate
C. Magnesium sulphate
D. Copper sulphate
E. None of the above/More than of the above

Ans. D

Exp: Refer to Que no 104

106. Bakelite is formed by the condensation of [62nd BPSC]

- A. Urea and formaldehyde
B. Phenol and formaldehyde
C. Phenol and acetaldehyde
D. Melamine and formaldehyde
E. None of the above/More than one of the above

Ans. B

Exp: Phenol and Formaldehyde form Bakelite by condensation reaction. It is used in making combs, electrical switch handles of utensils, and computer discs.

107. Biological preservative chiefly contains [Bihar Assistant Prosecution Officer]

- A. Ethyl Alcohol B. Formaldehyde
C. Ammonia D. Chlorine

Ans. B

Exp: Formaldehyde popularly known as Formalin (40%) solution is used to preserve biological specimens and to prepare Bakelite (a phenol-formaldehyde resin), urea-formaldehyde glues and other polymeric products.

CHEMISTRY IN EVERY DAY: FOOD PRESERVATIVES, SOAP AND DETERGENT, FERTILIZERS.

108. Which among the following is the main constituent of biogas? [67Re, 67 Cancelled BPSC]

- A. Propane B. Butane
C. Ethane D. Methane
E. None of the above/ more than one of the above

Ans. D.

Exp: Liquefaction of natural gas produces LPG. **Butane** is the main constituent of LPG, while smaller traces of propane and ethane are also present. **Ethyl Mercaptan** is added to the LPG which adds a strong odour and helps in leakage detection.

- ❖ **Butane**(C_4H_{10}) is a hydrocarbon and a highly flammable, colourless, odourless, easily liquefied gas. It is used as fuel in cigarette lighters and portable stoves, as a refrigerant and as a propellant in aerosols. Butane is also found in Liquefied Petroleum gas (LPG). Inhaling butane can cause 'sudden sniffing death'.
- ❖ Methane is the main component of biogas, CNG and Natural gas.
- ❖ **CNG** is natural gas compressed under high pressure; it consists of (80-90%) **Methane**, other gases are propane, ethane, nitrogen, carbon dioxide, hydrogen, etc.
- ❖ It is commonly known as green fuel because of its lead and sulphur-free character.
- ❖ **Biogas** contains up to 75% of methane. It burns without smoke and leaves no residue like ash in wood, or coal burning. Its heating capacity is high and also used for **lighting**.

109. Gobar gas contains mainly [30th Bihar Judicial Services]

- A. Methane B. Ethylene
C. Propylene D. Acetylene

Ans. A

Exp: Biogas when produced by the anaerobic decomposition of cow dung, it is called Gobar gas.

110. LPG used as domestic fuel chiefly contains [62nd BPSC]

- A. Methane B. Propane
C. Ethylene D. Butane

E. None of the above/More than one of the above
Ans. D

Exp: Refer to Que no 108

111. The major component of the LPG is [30th Bihar Judicial Services]

- A. Methane B. Butane
C. Ethane D. Propane

Ans. B

Exp: Refer to Que no 108

112. Which of the following gases is used in cigarette lighters? [59th BPSC]

- A. Butane B. Methane
C. Propane D. Radon

Ans. A

Exp: Refer to Que no 108

113. The major component of CNG is [66 Re BPSC]

- A. CO_2 B. N_2
C. H_2 D. CH_4
E. None of the above/More than one of the above

Ans. D

Exp: Refer to Que no 108

114. The main component of greenhouse gases is [65th BPSC]

- A. Carbon dioxide B. Methane
C. Nitrous oxide D. Ozone
E. None of the above/More than one of the above

Answer: E

Exp: The greenhouse effect is a process that occurs when gasses in Earth's atmosphere trap the Sun's heat. **Water vapour** (H_2O), **carbon**

dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), and ozone (O_3), are the Primary (major) greenhouse gasses.

BCW Bits:

The Kyoto Protocol is related to greenhouse gases.

❖ The enhanced greenhouse effect refers to the additional radiative forcing resulting from increased concentrations of greenhouse gasses induced by human activities. **The main greenhouse gases** whose concentrations are rising are **carbon dioxide, methane, nitrous oxide, hydrochlorofluorocarbons (HCFCs), hydrofluorocarbons (HFCs) and Ozone** in the lower atmosphere.

❖ The favourable Greenhouse effect is necessary to maintain a habitable temperature on Earth, without it the Earth will freeze. In the absence of the natural greenhouse effect, the surface of the Earth would be approximately 33°C cooler.

115. The main component of greenhouse gases is [64th BPSC]

- A. Carbon dioxide B. Methane
- C. Nitrous oxide D. Ozone
- E. None of the above/More than one of the above

Ans. E

Exp: Refer to Que no 114.

116. Which of the following is a greenhouse gas? [CDPO 2022 BPSC]

- A. Carbon Dioxide B. Ozone
- C. CFC D. Methane
- E. None of the above/More than one of the above

Ans. E

Exp: Refer to Que no 114.

117. Which gas contributes to global warming through its greenhouse effect? [68th BPSC Pre]

- A. Methane
- B. Carbon dioxide
- C. Nitrous oxide

D. More than one of the above

E. None of the above

Ans. D (A, B and C)

Exp: Refer to Que no 114

118. Which one of the following gases is considered an atmospheric pollutant? [CDPO 2018 BPSC]

- A. Oxygen
- B. Ozone
- C. Sulphur dioxide (SO_2)
- D. Nitrogen
- E. None of the above/More than one of the above

Ans. C

Exp: Sulphur dioxide is a colourless reactive air pollutant with a strong odour. The main source of sulphur dioxide is fossil fuel combustion and natural volcanic activity. This SO_2 combines with water vapour and forms sulphuric acid, which is the main constituent of acid rain.

119. The incomplete burning of petrol and diesel produces [66th BPSC]

- A. Nitric oxide
- B. Nitrogen dioxide
- C. Carbon dioxide
- D. Carbon monoxide
- E. None of the above/More than one of the above

Ans. D

Exp: The incomplete burning of petrol, diesel, charcoal grills, stoves etc produces carbon monoxide.

Carbon monoxide:

❖ It is one of the most serious air pollutants because when we inhale Carbon monoxide the oxygen in Red Blood Cells is replaced by carbon monoxide as CO binds with RBC more rapidly than oxygen and prevents oxygen from reaching tissues that lead to tissue damage, brain damage or even sudden death.

❖ Carbon Monoxide (CO) combines with Haemoglobin to form **carboxyhaemoglobin**. CO has 200–240 times more affinity for haemoglobin than oxygen..

- ❖ It is a colourless and odourless gas, highly poisonous to living beings because of its ability to block the delivery of oxygen to organs and tissues.
- ❖ It is produced as a result of incomplete combustion of carbon.
- ❖ Carbon monoxide is mainly released into the air by automobile exhaust. Other sources, which produce CO, involve incomplete combustion of coal, firewood, petrol, etc.

120. Inhalation of air polluted with carbon monoxide (CO) is dangerous because [CDPO 2022 BPSC]

- A. CO combines with oxygen in blood
- B. CO combines with haemoglobin
- C. CO removes water from the body and causes dehydration
- D. CO causes coagulation of proteins in the body
- E. None of the above/More than one of the above

Ans. B.

Exp: Refer to Que no 119.

121. The quality of petrol is expressed by [66 Re BPSC]

- A. Cetane number B. Octane number
- C. Butane number D. Reynolds number
- E. None of the above/More than one of the above

Ans. B

Exp: The quality of petrol is based on its Anti-knock quality expressed by **octane number** and that of diesel by **cetane number**.

Note: Lead (tetraethyl lead) is used as an anti-knock agent in fuels but has been banned due to its cancerous nature.

122. Detergents used for cleaning clothes are [63rd BPSC]

- A. Carbonates B. Bicarbonates
- C. Bismuthates D. Sulfonates
- E. None of these/More than one of these

Ans. D

Exp: Detergents are **sodium or potassium salts** of long-chain **benzene sulphonic acid**.

Detergent is salt of strong acid and strong base with high molecular weight, it is chemically called **surfactant**, with refining and cleaning properties it combines with dirt and impurities to make them soluble and does not form a scum in hard water also called soap less soap as it dissolves in both hard water and soft water.

- ❖ Detergent's aqueous solution is neutral and used for washing delicate clothes.
- ❖ Sodium silicate and sodium sulphate are mixed as a detergent in washing powder to keep the dryness and basicity of the powder.

BCW BITS:

- ❖ Emulsifiers are long-chain compounds with polar groups such as soap, sulphates, and sulphonic acid. Iodine also acts as an emulsifier.

The cleansing action of soaps and detergents is due to the formation of an oil-water emulsion on clothes, dishes, etc. The soap contains **hydrophilic (polar)** water-loving and **hydrophobic (non-polar)** water-repelling parts. The hydrophobic part gets attached to the dirt particle and the hydrophilic part gets attached to water. The cleansing action is due to the **hydrophobic part**.

- ❖ **Saponification** is the reaction by which soaps are formed, the reaction can be represented by

Ester + Alkali → Salt (soap) + Glycerol

- ❖ Soap forms scum when reacted with calcium and magnesium ions present in hard water, not suitable for washing clothes.

Ingredients that are added to soap to provide some peculiar characteristics-

- ❖ **Sodium Resonate** gives the foaming characteristic to the soap.
- ❖ **Glycerol** provides Smoothing characteristics
- ❖ **Sulphur** makes its anti-Acne.
- ❖ **Sodium silicate** makes the soap retain its shape.

123. Soap removes grease by [62nd BPSC]

- A. Coagulation B. Adsorption
- C. Emulsification D. Osmosis

- E. None of the above/More than one of the above

Ans. C

Exp: Refer to Que no 122

124. Detergents used for cleaning are [Bihar Assistant Prosecution Officer]

- A. Nitrates B. Carbonates
C. Sulphonates D. Bicarbonates

Ans. C

Exp: Refer to Que no 122

125. The liquified gas that is used in dry cleaning along with a suitable detergent is [AAO 2022 BPSC]

- A. Water gas B. CO₂
C. Petroleum gas D. NO₂
E. None of the above/more than one of the above

Ans. B

Exp: Tetrachloroethene (Cl₂C=CCl₂) is widely used in the dry cleaning of fabrics. It is also used for degreasing metal parts. Sometimes it is called "dry cleaning fluid". The problem associated with tetrachloroethene is its contaminating properties vis-a-vis groundwater. So, these days liquified CO₂ is being used as a dry cleaning agent to avoid groundwater pollution.

126. The elements present in urea are [66th BPSC]

- A. C, H, O B. C, N, O
C. C, H, N D. C, H, N, O
E. None of the above/More than one of the above

Ans. D

Exp: Urea, also known as carbamide, is an organic compound with the chemical formula CO(NH₂)₂. The components of urea are **one carbon, four hydrogens, two nitrogens, and one oxygen**. Urea serves an important role in the metabolism of nitrogen-containing compounds by animals and is the main nitrogen-containing substance in the urine of mammals. It is a colourless, odourless solid, highly soluble in water, and practically non-toxic. Dissolved in water, it is neither acidic nor alkaline.

- ❖ Urea has important uses as a fertilizer and feed supplement, as well as a starting material for the manufacture of plastics and drugs. Urea is now prepared commercially in vast amounts from liquid ammonia and liquid carbon dioxide.
- ❖ Urea was first isolated from urine in 1773 by the French chemist **Hilaire-Marin Rouelle**. Its preparation by the German chemist **Friedrich Wöhler** from ammonium cyanate in 1828 was the first generally accepted laboratory synthesis of a naturally occurring organic compound from inorganic materials.

127. Match List-I with List-II and select the correct Ans using the codes given below the lists: [62nd BPSC]

List-I [Substances]	List-II [Applications]
A. Aspartame	1. Synthetic rubber
B. Freon	2. Antihistamine
C. Neoprene	3. Artificial Sweetener
D. Benadryl	4. Refrigerant
A. a-1, b-3, c-4, d-2	
B. a-2, b-4, c-1, d-3	
C. a-3, b-4, c-1, d-2	
D. a-3, b-1, c-2, d-4	
E. None of the above/More than one of the above	

Ans. C

Exp: **Aspartame** is an artificial sweetener used in cold drinks and ice creams.

Neoprene, also called synthetic rubber, is a polymer of chloroprene.

Benadryl is used for allergies, rashes, itching, and the common cold.

Freon used as a refrigerant is a chlorofluorocarbon, an agent of global warming and the greenhouse effect.

128. Artificial sweetener, which is stable (non-decomposable) under cold conditions only, is [AAO 2022 BPSC]

- A. saccharine B. sucralose
C. alitame D. aspartame

- E. None of the above/More than one of the above

Ans. C

Exp: Note: the correct answer is Aspartame but according to BPSC answer key it's alitame

- ❖ **Aspartame** is the methyl ester of dipeptide formed from aspartic acid and phenylalanine. It is an artificial sweetener and 100 times sweeter than sugarcane. Aspartame is used in cold foods and soft drinks because it is unstable at cooking temperatures.
- ❖ **Alitame** is a high-potency sweetener, it is difficult to control the sweetness of food while using it and it is more stable than aspartame, hence is used at cooking temperature or high-temperature baking or cooking.
- ❖ **Sucralose**, a substituted disaccharide is a nonnutritive **sweetener** that is synthesized by selective chlorination of sucrose. It is also **highly stable** at elevated temperatures that are often used in food, beverage, and drug manufacturing processes so that product sweetness levels can be maintained following cooking, baking, and pasteurization.
- ❖ Sucralose also has **excellent stability** in low-pH products so sweetness degradation is not a determining factor in the shelf-life of such products.
- ❖ By weight, sucralose is about 600 times sweeter than sucrose.

129. The artificial sweetener containing chlorine that has the appearance and taste as that of sugar and remains stable at cooking temperature, is [CDPO 2018 BPSC]

- A. Aspartame B. Saccharine
C. Sucralose D. Alitame
E. None of the above/More than one of the above

Ans. C

Exp: Refer to Que no 128

130. Iodised salt contains [62nd BPSC]

- A. Free iodine B. Calcium iodide
C. Magnesium iodide D. Potassium iodide

- E. None of the above/More than one of the above

Ans. D

Exp: To iodise normal salt, it is sprinkled with potassium iodide or sodium iodide. Iodized salt boosts thyroid function.

Iodine is the only non-metal which has lustre.

Note: Deficiency of Iodine causes Goitre.

131. The gas used in the manufacture of Vanaspati Ghee from Vanaspati Oil is [62nd BPSC]

- A. Hydrogen B. Oxygen
C. Nitrogen D. Carbon dioxide
E. None of the above/More than one of the above

Ans. A

Exp: Hydrogenation of vegetable oils using **nickel** as a catalyst gives edible fats (margarine and vanaspati ghee).

Note: A Catalyst does not take part in the reaction; it only alters the rate of the reaction.

132. Which of the following is used in Brewing of beer [AAO 2022 BPSC]

- A. Penicillin B. Yeast
C. Club fungi D. Mushroom
E. None of the above more than one of the above

Ans. B

Exp: Yeasts are unicellular organisms. Yeasts have two main uses in food production:

Baking and making alcoholic beverages. Yeast is **fungi**. It converts its food sugar and starch through **fermentation**, into carbon dioxide and alcohol.

Note: Mushrooms are also fungi and are edible.

133. The breath test conducted by police to check drunken drivers has which one of the following on the filter paper? [66th BPSC]

- A. Potassium dichromate-sulfuric acid
B. Potassium permanganate-sulfuric acid
C. Silica gel coated with silver nitrate
D. Turmeric

- E. None of the above/More than one of the above

Ans. A

Exp: A **Breathalyzer** includes a mouthpiece and two chambers containing liquid connected to a meter that detects a colour change. The chamber contains several compounds such as **Potassium dichromate**, **Sulfuric acid**, and **Silver nitrate**.

To use the Breathalyzer, the subject exhales through the mouthpiece into a test chamber filled with a reddish-orange solution of **potassium dichromate** ($K_2Cr_2O_7$). The alcohol reacts with the reddish-orange potassium dichromate solution and turns **green**. The degree of colour change is directly related to the level of alcohol in the expelled air.

Note: The more ethanol is present, the greater the amount of green colour produced. It includes both oxidation and reduction reactions.

134. A 'Breath Test' used by traffic police to check drunken driving case uses [Sanitary Waste Management BPSC]

- A. Turmeric on filter paper
- B. Silica gel coated with silver nitrate
- C. Potassium permanganate-sulphuric acid
- D. Potassium dichromate- sulphuric acid

Ans. D

Exp: Refer to Que no 133.

135. Select the incorrect statement out of the following. [67th Cancelled BPSC]

- A. Cotton is suitable for use as clothing in summer because it absorbs moisture.
- B. Polycarbonate is used for making CDs.
- C. Acrylic is also called artificial silk as it is prepared from cotton but has to shine like silk.
- D. Teflon is used for coating non-stick kitchen wares.
- E. None of the above/more than one of the above

Ans. C

Exp: **Rayon** is a man-made fibre prepared from a natural raw material called cellulose by chemical treatment. It is chemically identical to cotton but

it shines like silk. Since rayon resembles silk in appearance therefore **rayon is also called artificial silk**.

Hence option C is incorrect.

BCW Bits:

- ❖ Non-stick cookware is coated with Teflon (poly tetra fluoro Ethylene), and the fluorine present repels almost all other materials preventing them from sticking.
- ❖ The first synthetic fibre was Nylon made in 1931.
- ❖ The monomer of nylon was caprolactam.

136. The first synthetic fibre prepared was [Bihar Assistant Prosecution Officer]

- A. Rayon
- B. Nylon
- C. Terycott /Tricot
- D. Polyester

Ans. B

Exp: The first fully synthetic fibre prepared was Nylon. In 1931, it was prepared without using any natural raw material. It was made from coal, water and air. It was strong, elastic and light and also lustrous and easy to wash. It was used to make clothes. Many articles made of Nylon includes socks, ropes, tents, toothbrushes, car seat belts, sleeping bags, curtains, and parachutes. A nylon thread is stronger than a steel wire.

137. The process of adding Sulphur to heated rubber is known as [Bihar Assistant Prosecution Officer]

- A. Sulfonation
- B. Sulfuration
- C. Vulcanization
- D. Sulfation

Ans. C

Exp: The process of adding sulfur to heated rubber is known as the Vulcanization of Rubber. In this process, a mixture of raw rubber with sulphur and an appropriate additive is heated at a temperature range between 373 K to 415 K. On vulcanization, sulphur forms cross links at the reactive sites of double bonds and thus rubber gets stiffened.

The rubber (natural) is soft at high temperatures (>335 K) and brittle at low temperatures (<283 K) and shows absorption capacity. It is soluble in non-polar solvents and is non-resistant to attack by oxidizing agents. To improve the physical

properties the process of vulcanization is done.

5% of sulphur is used as a cross linking agent in the manufacturing of tyre rubber.

138. Non-stick cooking utensils are coated with a layer of [Bihar Assistant Prosecution Officer]

- A. Polystyrene B. PVC
C. Black Paint D. Teflon

Ans. D

Exp: Non-stick cooking utensils are coated with a layer of Teflon.

BCW Bits:

Polymer: The word polymer is derived from two Greek words: "Poly—> many & mer—> unit".

It is defined as high molecular mass macromolecules, consisting of repeating structural units derived from the corresponding monomers. The process of formation of polymers from respective monomers is called polymerization

(Type: 1. Addition or chain growth polymerization; type: 2. Condensation or step growth polymerization)

It is derived from natural sources (Natural polymers- proteins, cellulose, starch, rubber) as well as synthetic sources (synthetic polymers- plastic (polythene), synthetic fibre (Nylon 6,6), synthetic rubber (Buna-S)).

❖ **Biodegradable Polymers:** Designed and Developed to address the problem of polymeric solid wastes as they are resistant to environmental degradation. These polymers contain functional groups similar to the functional groups present in biopolymers. Aliphatic polyesters are one of the important classes of biodegradable polymers.

Examples: -

1. **PHBV:** Poly beta-hydroxybutyrate– co-beta-hydroxy valerate. It is obtained by the copolymerization (more than one monomeric species is allowed to polymerize) of 3- hydroxybutanoic acid and 3- hydroxypentanoic acid. It is used in specialty packaging, orthopaedic devices and in controlled release of drugs.
2. **Nylon 2-Nylon-6:** - it is an alternating copolymer of glycine ($\text{H}_2\text{N}-\text{CH}_2-\text{COOH}$) and amino caproic acid ($\text{H}_2\text{N}(\text{CH}_2)_5\text{COOH}$).

Important polymers:

Polymer	Monomer	Uses
polypropene	propene	Ropes, toys, pipes, fibres, etc.
polystyrene	Styrene	Insulator, wrapping material, toys, radio and television cabinet.
Polyvinyl chloride (PVC)	Vinyl chloride	Raincoats, handbags, vinyl flooring, water pipes.
Urea-formaldehyde Resin	❖ Urea ❖ Formaldehyde	Unbreakable cups and laminated sheets
Glyptal	❖ Ethylene glycol ❖ Phthalic acid	Paints and lacquers.
Bakelite	❖ Phenol ❖ Formaldehyde	Combs, electrical switches, handles of utensils and computer discs.
Kevlar	polyamides	Bullet proof jackets.
Natural Rubber	Isoprene	Medical devices, surgical gloves, tires etc.
Polyethene terephthalate	Ester	Responsible for the essence of fruits
Teflon	Tetrafluoroethylene	Non-stick pan

Test Your Knowledge

1. Which among the following is not matched correctly:

- A. Philosopher's wool $\rightarrow$ ZnO.
 B. Aqua Regia $\rightarrow$ $[3\text{HCl} + \text{HNO}_3]$.
 C. Plaster of Paris $\rightarrow$ $\text{CuSO}_4 \cdot 2\text{H}_2\text{O}$.
 D. More than one of the above
 E. None of the above.

Ans. C

2. Which among the following polymers is /are biodegradable?

- A. PHBV
 B. Nylon 2-Nylon-6
 C. Glyptal
 D. More than one of the above
 E. None of the above.

Ans. D (a and b)

3. Yellow cake which is smuggled across the borders represents

- A. Heroin
 B. Cocaine
 C. Uranium Oxide
 D. More than one of the above
 E. None of the above

Ans. C

4. Consider the following statements

- (i) Bromine is the only non-metal to exist in liquid form at room temperature.
 (ii) Krypton is the most abundant Noble gas found in the atmosphere.
 (iii) Oxygen is the most abundant element in the Human body.
 (iv) Most of the metals are oxidising agents.
 Which of the following statements are correct?

- A. (i) and (ii) only
 B. (ii) and (iii) only
 C. (i) and (iii) only
 D. More than one of the above
 E. None of the above

Ans. C

MISCELLANEOUS

139. The chemical formula for heavy water is [66 Re BPSC]

- A. H_2O B. N_2O
 C. D_2O D. CuO
 E. None of the above/More than one of the above

Ans. C

Exp: Deuterium is an isotope of hydrogen. It reacts with oxygen to form D_2O . It is used as a moderator in reactors to slow down neutrons produced by fission to sustain the chain reaction.

140. Heavy water is [Bihar Assistant Prosecution Officer]

- A. Tap Water B. Lake Water
 C. Deuterium Oxide D. River Water

Ans. C

Exp: Refer to Q.No 139

141. Which of the following is a commonly used moderator in a nuclear reactor? [Sanitary Waste Management BPSC]

- A. Heavy water B. Graphite
 C. Water D. All of the above

Ans. D

Exp: Graphite, Heavy Water (D_2O), and Water are the three most commonly used moderators. This slows down fast neutrons and aids in keeping a nuclear chain reaction going.

142. In a nuclear reactor, which one of the following is used as a fuel? [30th Bihar Judicial Services]

- A. Coal B. Uranium
 C. Radium D. Diesel

Ans. B

Exp: Uranium is used as a fuel in a nuclear reactor. Nuclear reactors are used to generate nuclear energy by a process called nuclear fission.

❖ In this process, a heavy metal such as Uranium, plutonium or thorium is bombarded with low-energy neutrons. The larger nuclei then split into two or more smaller or

lighter nuclei. During this reaction, a large amount of energy is released which can be used to produce steam and further generate electricity. In nuclear reactors designed for electric power generation, controlled and self-sustaining chain reactions take place and energy is also released at a controlled rate.

143. Which among the following would cause the bright red colour due to the bursting of crackers? [67 Re BPSC]

- A. Sodium B. Sulphur
C. Magnesium D. Strontium
E. None of the above/ more than one of the above

Ans. D.

Exp: From the table, it is clear that the bright red colour is due to strontium.

Metal Salt	Colour in firework
Strontium Carbonate	Red
Sodium Nitrate	Yellow
Barium Chloride	Green
Copper Chloride	Blue

144. The highest viscosity among the following is of [66th BPSC]

- A. Water B. Air
C. Blood D. Honey
E. None of the above/More than one of the above

Ans. D

Exp: **Viscosity** is the property of fluids wherein an internal frictional force (similar to one when one walks on a solid surface) comes into play when the fluid is in motion in the form of layers. The SI unit of viscosity is Pa Sec (Pascal Second) and the SI unit of dynamic viscosity is **poiseuille (Pl)**. It is measured using a **viscometer**.

BCW BITS: -

- ❖ Of all fluids, gases have the lowest viscosities, and thick liquids have the highest. The viscosity of liquids decreases rapidly with an increase in temperature, and the viscosity of gases increases with an increase in temperature. Thus, upon heating, liquids

flow more easily, whereas gases flow more sluggishly.

- ❖ Generally, thin liquids like water, alcohol, etc. are less viscous than thick liquids like coal tar, glycerin, honey, etc.

145. To protect broken bones, plaster of Paris is used. It is [63rd BPSC]

- A. Slaked lime B. Calcium carbonate
C. Calcium oxide D. Gypsum
E. None of the above/More than one of the above

Ans. D

Exp: Plaster of Paris is prepared by heating gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) at 100°C or 373K . On heating, it loses three-fourths of its water molecules and forms the plaster of Paris.

Plaster of Paris is calcium sulphate hemihydrate ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$). It consists of white powder and sets into a hard mass when pasted with water and allowed to dry. The setting of plaster of Paris is catalysed by **NaCl**, while reduced by **alum or borax**.

146. What is Particulate matter? [62nd BPSC]

- A. Solid residue B. Air pollutant
C. Water pollutant D. Soil pollutant
E. None of the above/More than one of the above

Ans. B

Exp: Solid and liquid particles which are suspended in air are called particulate matter. It includes both organic and non-organic materials. E.g. dust, smoke, pollen, liquid droplets, PM_{10} , $\text{PM}_{2.5}$ (it is more dangerous).

Particulate matter helps in cloud formation which is why artificial particulate matter like dry ice (solid carbon dioxide), potassium iodide, sodium iodide, and salt powder is used for artificial rain.

The positive impact of particulate matter is the precipitation while the negative effect is acid rain.

Note: The Central Government has launched the National Clean Air Programme to tackle the air pollution problem across the country in a comprehensive manner with targets to achieve a 20% to 30% reduction in Particulate Matter concentrations by 2024.

147. Photochemical smog is a resultant of the reaction among [UPSC 2013]

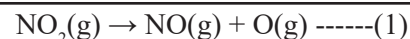
- A. NO_2 , O_3 and Peroxyacetyl Nitrate in The Presence of Sunlight
- B. CO , O_2 and Peroxyacetyl Nitrate in The Presence of Sunlight
- C. CO , CO_2 and NO_2 at Low Temperature
- D. High Concentration of NO_2 , O_3 And CO in the Evening

Ans. A

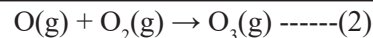
Exp: Smog is an example of Air Pollution. The word 'Smog' is derived from the two words Smoke and Fog. Smog has two types; one is classical smog and another is photochemical smog.

BCW BITS:

- ❖ **Classical smog:** This type of smog occurs in cool humid climates. It is a mixture of smoke, fog and sulfur dioxide. It is a reducing mixture so chemically it is also known as reducing smog.
- ❖ **Photochemical smog:** This type of smog is common in warm, dry and sunny climates. The main components (ozone, nitric oxide, acrolein, formaldehyde and Peroxyacetyl nitrate (PAN)) of the photochemical smog result from the action of sunlight on unsaturated hydrocarbons and Nitrogen oxides produced by automobiles and factories. Photochemical smog consists of a high concentration of oxidizing agents and is, therefore, called oxidizing smog.
- ❖ **Formation of Photochemical smog:**
 - When fossil fuels are burnt, a variety of pollutants are emitted into the troposphere. Two of these emitted pollutants are Hydrocarbons (Unburnt fuel) and nitric oxide (NO). When their emission reaches a certain higher level, a chain reaction occurs as they interact with the sunlight in which NO is converted into NO_2 (nitrogen dioxide). This NO_2 absorbs energy from sunlight and breaks up into nitric oxide and free oxygen atoms.



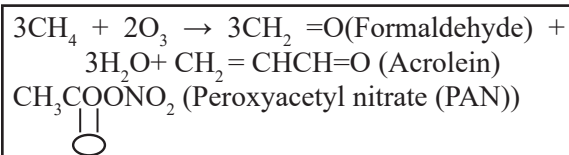
Oxygen atoms are very reactive and combine with the O_2 in the air to produce ozone.



The ozone thus formed reacts rapidly with the $\text{NO}(\text{g})$ (formed in reaction 1) to regenerate NO_2 . NO_2 is a brown-colored gas and when it reaches a sufficiently high level, it contributes to Haze.



Ozone is a toxic gas and both NO_2 and O_3 are strong oxidizing agents and can react with the unburnt hydrocarbons in the polluted air to produce chemicals such as formaldehyde, acrolein and peroxyacetyl nitrate (PAN).



❖ Effects of photochemical smog:

- Ozone and PAN cause eye irritations.
- O_3 and NO cause nose and throat irritation and when exposed to their higher concentration, it can cause headache, chest pain, dryness of the throat, cough and difficulty in breathing.
- It leads to the cracking of rubbers and damages plants as well.
- It also causes corrosion of metals, stones, building materials, rubber and painted surfaces.

❖ Ways to control photochemical smog:

- Primary precursors such as NO_2 and hydrocarbons can be controlled by using catalytic converters in automobiles as it will prevent their release into the atmosphere.
- Plantation of → Pinus, Juniparus, Quercus, Pyrus and Vitis are useful in reducing the concentration of Nitrogen Oxide.

Hence, option A is the correct answer.

148. The Bhopal Gas Tragedy was caused by the leakage of which gas? [30th Bihar Judicial Services]

- A. Phosgene B. Carbon dioxide
C. Methyl isocyanate D. Chlorine

Ans. C

Exp: The Bhopal Gas Tragedy was caused by the leakage of Methyl isocyanate gas. It is considered the world's worst industrial tragedy. At midnight on **2 December 1984, Methyl isocyanate (MIC)** started leaking from the **Union Carbide plant** (an American pesticide company). The gas was highly poisonous and within three days caused the death of more than 8,000 people. Most of the people who were exposed to this gas came from working-class families. 50,000 people became so sick that they were not even able to work. The survivors of this tragedy developed severe respiratory disorders, eye problems and other disorders. Children were also severely affected and developed peculiar abnormalities

149. 'God Particle' is [62nd BPSC]

- A. Neutrino B. Higgs Boson
C. Meson D. Positron
E. None of the above/More than one of the above

Ans. B

Exp: The **Higgs Boson** was proposed in 1964 by **Peter Higgs** and four other scientists. It is a **subatomic particle** associated with the Higgs field. Scientists confirmed the existence of the Higgs boson in 2012 through the Large Hadron Collider experiment at CERN Switzerland.

"The Higgs boson is also called the '**God Particle**'.

150. Cosmic rays [59th BPSC]

- A. are charged particles
B. are uncharged particles
C. can be charged as well as uncharged
D. None of the above

Ans. A

Exp: Austrian physicist **Victor Hess** has discovered cosmic rays. **Cosmic rays** are **high-energy particles** that move through space at nearly

the **speed of light**. These high-energy particles **arriving from outer space** are mainly (89%) protons- nuclei of hydrogen, the lightest and most common element in the universe but they also include nuclei of helium (10%) and heavier nuclei (1%). When they arrive at Earth, they collide with the nuclei of atoms in the upper atmosphere, creating more particles.

Note The origin of cosmic rays is not defined completely, but most scientists believe that they originate from "supernovas".

151. Water gas is a mixture of [CDPO 2022 BPSC]

- A. $\text{CO} + \text{H}_2$ B. $\text{CO} + \text{O}_2$
C. $\text{CO} + \text{N}_2$ D. $\text{CO}_2 + \text{H}_2$
E. None of the above/More than one of the above

Ans. A

Exp: Water gas is a mixture of carbon monoxide and hydrogen gas. The mixture of CO and H_2 is used for the synthesis of methanol and the number of hydrocarbons hence it is also called **synthesis gas or syngas**.

Note: $\text{CO} + \text{H}_2 + \text{N}_2 =$ Producer Gas; Producer gas is low calorific fuel that is produced by passing air and steam through the burning bed of biomass like coal. It is also called "**Airgas**".

152. Substances that readily absorb moisture from other substances without chemically reacting with them are called [CDPO 2022 BPSC]

- A. Desiccants
B. Deliquescent substances
C. Efflorescent substances
D. Dehydrating agents
E. None of the above/More than one of the above

Ans. A

Exp: A desiccant is a hygroscopic (phenomenon of attracting and holding water molecules) material used to maintain a state of dryness.

Desiccants are commonly used to keep products dry and stable. Dry desiccants can absorb moisture from the air either by physical adsorption or by chemical reaction, and thus reduce the humidity in the headspace of sealed containers.

Desiccants are often incorporated into packaging for moisture-sensitive products.

Silica Gel, molecular sieves, and clay minerals are all common desiccating materials used.

153. Name the scientist who stated that matter can be converted into energy. [CDPO 2018 BPSC]

- A. Boyle B. Lavoisier
C. Avogadro D. Einstein
E. None of the above/More than one of the above

Ans. D

Exp: In 1905, **Einstein** developed the special theory of relativity. One of the implications of this theory was that matter and energy are interchangeable with one another.

It states that a mass (m) can be converted into an amount of energy (E) [$E=mc^2$], where c is the speed of light. Because the speed of light is a large number and thus c squared is huge, a small amount of matter can be converted into a tremendous amount of energy. This equation is the key to the power of nuclear weapons and nuclear reactors.

154. _____ is widely seen as the mass Fuel of the Future [District Art and Culture BPSC]

- A. Oxygen B. Nitrogen
C. Hydrogen D. Ozone
E. None of the above/more than one of the above.

Ans. C.

Exp: Hydrogen, the **most abundant element** in the universe and the third most abundant on the surface of the globe, is being visualized as the major future source of energy. Hydrogen is an energy carrier that can be used in internal combustion engines or fuel cells producing virtually no greenhouse gas emissions when combusted with oxygen. The only significant **emission is water vapour.**

Liquid nitrogen is capable of making an engine run without carbon emissions. However, car manufacturers are still studying the efficiency and practicality of this option. One of the most

pressing matters is the cost, especially since the infrastructure is currently limited.

155. Plastic bags containing potato chips are filled with an antioxidant, known as [Sanitary Waste Management BPSC]

- A. Oxygen B. Nitrogen
C. Hydrogen D. Carbon monoxide

Ans. B

Exp: When fats and oils are oxidized, they become rancid and their smell and taste change. Usually, substances which prevent oxidation (antioxidants) are added to foods containing fats and oil. Thus, Plastic bags containing potato chips are filled with nitrogen gas.

156. In India, the use of carbofuran, methyl parathion, phorate and triazophos is viewed with apprehension. These chemicals are used as [UPSC 2019]

- A. Pesticides in agriculture
B. Preservatives in processed foods
C. Fruit-ripening agents
D. Moisturizing agents in cosmetics

Ans. A

Exp: **Carbofuran, Methyl Parathion, Phorate** and **Triazophos** are used as Pesticides in agriculture. Pesticides are used to protect crops against insects, weeds, fungi and other pests. These chemicals are toxic to humans and they can cause acute as well as chronic health effects depending on the amount and ways in which a person is exposed to it.

157. Which of the following is not biodegradable? [29th Bihar Judicial Service]

- A. Domestic sewage B. Lab detergents
C. Soap D. Plant leaves

Ans. B

Exp: Substances that are broken down by biological processes are known as biodegradable substances. And Substances that are not broken down by biological processes are known as non-biodegradable substances.

Synthetic detergents or Lab detergents are cleansing agents having all the properties of soap,

but do not contain any soap. It contains highly branched hydrocarbon chains which cannot be degraded easily and hence, causes environmental pollution.

Test Your Knowledge

1. Which of the following elements was called “eka-Aluminium” by Mendeleev?

- A. Scandium
- B. Gallium
- C. Germanium
- D. More than one of the above
- E. None of the above

Ans. B

2. Consider the following statements

- (i) Nitroglycerine is known as Noble's Oil
- (ii) RDX is known as Plastic Explosives
- (iii) Amatol is a mixture of TNT and Ammonium Nitrate
- (iv) RDX is an explosive that can't be shaped or moulded.

Which of the following statements are correct?

- A. (i) only
- B. (ii) and (iv) only
- C. (i), (ii) and (iii) only
- D. More than one of the above
- E. None of the above

Ans. C

3. Which of the following gases can be used for the storage of fresh samples of oil for a long time?

- A. Carbon dioxide or oxygen
- B. Helium or nitrogen
- C. Carbon dioxide or helium
- D. More than one of the above
- E. None of the above

Ans. B

4. Which of the following is not an ore of Iron?

- A. Bauxite

- B. Malachite
- C. Magnetite
- D. More than one of the above
- E. None of the above

Ans. D (A and B)

5. The relationship between Volume - Temperature is given by.

- A. Charles's Law
- B. Boyle's Law
- C. Avogadro Law
- D. More than one of the above
- E. None of the Above

Ans. A

6. When any metal is mixed with mercury to form an alloy then the resultant alloy is called-

- A. Homogeneous Alloy
- B. Amalgam
- C. Metalloid
- D. More than one of the above
- E. None of the above

Ans. B

7. Which among the following is/are physical changes?

- A. Evaporation of petrol
- B. Burning of Liquefied Petroleum Gas (LPG)
- C. Heating of an iron rod to red hot.
- D. More than one of the above
- E. None of the above

Ans. D (A & C)

8. Consider the following statements:

- i. The human body works within the pH range of 7.0 to 7.8.
- ii. When the pH of rainwater is less than 5.6, it is called acid rain.
- iii. Tooth decay starts when the pH value of the mouth increases.

Which of the statements given above is/are correct?

- A. i and ii only

- B. ii and iii only
- C. iii only
- D. More than one of the above
- E. None of the above

Ans. A

9. Which of the following statements is/are correct?

- (1) Boiling is a bulk phenomenon
- (2) Evaporation is a surface phenomenon
- (3) Increase in the pressure around the surroundings increases the boiling point of liquid
- (4) Increase in the pressure around the surroundings decreases the boiling point of the liquid.

- A. 1 only
- B. 2, 3 and 4 only
- C. 1, 2, and 3 only
- D. More than one of the above
- E. None of the above

Ans. C

10. Which of the following statements about benzene is INCORRECT?

- A. C_6H_6 is the molecular formula.
- B. When heated in air, benzene burns with sooty flame producing CO_2 & H_2O
- C. In Benzene the number of sigma bonds is 3 and the number of pi bonds is 12.
- D. More than one of the above
- E. None of the above

Ans. C

11. Glucose ($C_6H_{12}O_6$) can be prepared from which of the following

- A. Fructose
- B. Starch
- C. Cane Sugar
- D. more than one of the above
- E. None of the above

Ans. D (B and C)

12. Tincture of Iodine is

- A. 1-2 per cent of Iodine solution in an alcohol-water mixture
- B. 2-3 per cent of Iodine solution in an alcohol-water mixture
- C. 4-5 per cent of Iodine solution in an alcohol-water mixture
- D. More than one of the above
- E. None of the above

Ans. B

13. Which of the following statements is/are incorrect?

- A. On earth, escape speed is around 11.2km/s
- B. If a body is projected with a velocity greater than the escape speed then the path taken by it will be parabolic.
- C. The escape speed for a body is independent of the angle with which it is projected.
- D. More than one of the above
- E. None of the above

Ans. B

14. Which among the following is not an ideal condition for the formation of dew?

- A. Cloudy sky
- B. High relative humidity
- C. Calm air
- D. More than one of the above
- E. None of the above

Ans. A

15. Rate of diffusion of a gas is.

- A. Inversely proportional to the square root of its molecular mass.
- B. Inversely proportional to its molecular mass
- C. Directly proportional to the square of its molecular mass.
- D. More than one of the above
- E. None of the above

Ans. A



ONE LINER

Structure of Atoms

Dalton's Atomic theory: (given by John Dalton in 1808).

Postulates:

All matter is made of very tiny particles called atoms.

- ❖ Atoms are **indivisible particles**, which cannot be created or destroyed in a chemical reaction.
- ❖ Atoms of a given element are identical in mass and chemical properties.
- ❖ Atoms of different elements have different masses and chemical properties.
- ❖ Atoms combine in the ratio of small whole numbers to form compounds.
- ❖ The relative number and kinds of atoms are constant in a given compound.

Drawbacks:

- ❖ The **indivisibility of an atom was proved wrong**: an atom can be further subdivided into **protons, neutrons and electrons**. However an atom is the smallest particle that takes part in chemical reactions.

Fundamental particles of Atom:

Name	Discovered by	Atomic charge/C	Atomic mass/kg	Atomic mass/u	Relative charge	Relative mass
Electrons (e)	J J Thomson (1897)	$-1.602176 \times 10^{-19}$	9.109382×10^{-31}	0.00054	-1	0
Protons (p)	Goldstein (1886, as anode rays), Named by Rutherford (1919).	$+1.602176 \times 10^{-19}$	$1.6726216 \times 10^{-27}$	1.00727	+1	0
Neutrons (n)	James Chadwick (1932)	0	1.674927×10^{-27}	1.00867	0	1

- ❖ Atomic Nucleus → discovered by **Rutherford** in 1911, Contains **Nucleons (protons and neutrons)**.
- ❖ Native elements → contains single type of Atom,
- ❖ Compound → contains two or more types of Atoms.
- ❖ Deuteron → contains a proton and a neutron, (no electron), thus not a subatomic particle.
- ❖ Antiparticles → Electron - Positron.
- ❖ God particle → Higgs Boson.
- ❖ Father of Modern chemistry → Antonio Lavoisier.

Atomic Number (Z)	Mass Number (A)	Atomic Weight
Atomic Number = number of protons in the nucleus.	Mass Number = number of protons + number of neutrons.	The atomic weight is approximately equal to the Mass Number of an atom.

Isoelectronic	Isotopes	Isobars	Isotones
These are ions/atoms having equal number of electrons.	Atoms of an element with same atomic number but different atomic mass.	Atoms of different elements with Same atomic mass but different atomic numbers.	Atoms of two different elements with the same number of neutrons.

- ❖ Radioactivity, a nuclear reaction, was discovered by **Henry Bacquerel** in Uranium.
- ❖ Alpha, Beta and Gamma rays are emitted in a radioactive process.
- ❖ Radioactivity measured by → **Geiger Counter**.

Alpha rays	Beta rays	Gamma rays
❖ 'α' or He^{2+} or $(\text{He}^4)^{2+}$ ❖ Positively charged ❖ Velocity = 1/10 of velocity of light. ❖ High ionizing power.	❖ 'β' or $-1e$ ❖ Negatively charged ❖ Velocity = Velocity of Light. ❖ Ionizing power 100 times lesser than Alpha rays.	❖ 'γ' ❖ Neutral electro radiations. ❖ Velocity = Velocity of Light. ❖ Ionizing power 100 times lesser than Beta rays

Physical and Chemical Changes

Physical changes	Chemical changes
❖ Physical properties such as shape, size, colour, etc. changes while chemical composition remains the same ❖ It is a reversible process. ❖ Little or no transfer of heat energy.	❖ Both physical and chemical properties change, formation of new substances takes place. ❖ It is an irreversible process. ❖ Chemical change can be either endothermic or exothermic.

- ❖ Some substances can be obtained in pure state from their solutions by **crystallization**.
- ❖ **Corrosion** → The surface of some metals, such as iron, is corroded when they are exposed to moist air for a long period of time. Eg : Black coating on silver, Green coating on copper, Brown coating on Iron (Rusting).
- ❖ The rusting of iron can be prevented by painting, oiling, greasing, galvanizing, chrome plating, anodising or making alloys.
- ❖ **Galvanization** is a method of protecting steel and iron from rusting by coating them with a thin layer of **zinc**.

Solution

- ❖ Solution = Solute + Solvent.

Types of Solution :

- ❖ Gaseous solutions → Only gas can be dissolved in a gaseous solvent. eg: air (O_2 + other gasses dissolved in N_2).
- ❖ Liquid solutions → Gases, Liquids and Solids can be dissolved in a Liquid solvent.
- ❖ Solid solutions → A Solid solvent can also dissolve all three (solid, liquid and gas) types of solute.
- ❖ Dilute solution → Solute level did not reach the saturation point (i.e) more solutes can be dissolved in the solution.
- ❖ Concentrated solution → more solute is present than that can be dissolved.

- ❖ Saturated solution → no more solute can be dissolved in the solution.
- Solubility → moles of solute/ 100gm of solvent.
- Osmosis → process of movement of solvent towards the region with higher solute concentration from a region of lower solute concentration.

Application of osmosis:

- ❖ Water movement from soil into plant roots and subsequently into the upper portion of the plant is partly due to osmosis.
- ❖ The preservation of meat by salting and of fruits by adding sugar protects against bacterial action.
- ❖ Through the process of osmosis, a bacterium on salted meat or candid fruit loses water, shrivels and dies.
- Reverse Osmosis: The direction of osmosis can be reversed if a pressure larger than the osmotic pressure is applied at the solution side through a semipermeable membrane. This process is known as reverse osmosis.
- Pasteurization: The milk is heated to about 70°C for 15 to 30 seconds and then suddenly chilled and stored. By doing so, it prevents the growth of microbes. This process was discovered by **Louis Pasteur**.
- Boiling point of a liquid increases when impurities are added to it.
- Emulsion → Colloids of two or more immiscible liquids,

Types : **Oil in water** & **Water in oil**. For Ex: Milk (oil in water).

- Honey is not a colloid.
- Distillation → used for purification of liquids
- Chromatography → separates different components in a liquid mixture.
- **Solid camphor, Naphthalene, Dry ice** gets converted into their vapor form due to **Sublimation**.
- Antifreeze - Substances such as **Ethylene Glycol** are used as an antifreeze in cars

during winters, as they are added with water to lower its freezing point.

Inorganic Chemistry

The study of periodic relationships between the elements was started by **Johann Dobereiner**, but **Dmitri Mendeleev** was the first one to publish 'Periodic Law'.

Origin of Periodic Classification:

1. **German chemist Johann Dobereiner** → **Law of triad** → "the middle element of each of the Triads had an atomic weight about half way between the atomic weights of the other two."
2. **French geologist, A.E.B. de Chancourtois** → **Cylindrical table of elements** → Didn't get much attention.
3. **English chemist, John Alexander Newlands** → **Law of Octaves** → "Every eighth element has properties similar to the first element."
4. **German chemist, Lothar Meyer (1830-1895)** → Lothar Meyer plotted the physical properties such as atomic volume, melting point and boiling point against atomic weight and obtained a periodically repeated pattern. Lothar Meyer had developed a table of the elements that closely resembles the Modern Periodic Table.
5. **Russian chemist, Dmitri Mendeleev (1834-1907)** → He is generally credited with the development of the Modern Periodic Table. Periodic Law given by him – "The properties of the elements are a periodic function of their atomic weights."
6. Both Meyer and Mendaleev, in 1869 proposed that on arranging elements in the increasing order of their atomic weights, similarities appear in physical and chemical properties at regular intervals.

Mendeleev's periodic table

- ❖ Arranged elements in increasing order of atomic mass.
- ❖ Horizontal rows → Period (total 6 periods)
- ❖ Vertical columns → Groups (total 8 groups)

- ❖ Left blank spaces for elements not discovered till then, such as Eka-Aluminium and Eka-Silicon, also predicted their existence, which was later discovered as gallium and germanium.

Drawbacks

- ❖ Placed heavier elements before lighter ones.
- ❖ No place for different isotopes of an element.
- ❖ It was felt that elements should be classified on the basis of a more fundamental property rather than the atomic mass.

Henry Moseley: Modern Periodic Table

- ❖ Elements were arranged by their atomic number.

Modern Periodic Law: "The physical and chemical properties of the elements are periodic functions of their atomic numbers."

- ❖ Period -7, Group -18.
- ❖ Elements of the same group has similar valence shell electronic configuration and exhibits similar chemical properties.
- ❖ Elements of the same period reflects increasing number of electrons from left to right, and has different Valencies.
- ❖ The Valency is either equal to the number of electrons in the outermost orbit or 8 minus this number.

- ❖ **Valency** - Measure of combining capacity to form a covalent bond with an atom (or with how many no. of other atom will the concerned atom combine)
- ❖ Metals –78% of the known elements.
- ❖ Non-metals – located at the top of the periodic table, are less than 20 in number.
- ❖ Metalloids or Semi-Metals–Elements which lie at the border line between metals and nonmetals (e.g., Si, Ge, As).
- ❖ The reactivity on the left extreme of a period is high, because these element can lose electrons with ease (has low ionization enthalpy).
- ❖ Highly reactive elements do not occur in nature in a free state; they usually occur in the combined form.
- ❖ **Cu (Copper), Ag (Silver), Fe (Iron)** are found in both free and combined states.
- ❖ Oxides formed of the elements on the left of the periodic table → **Basic**.
- ❖ Oxides of the elements on the right of the periodic table → **Acidic**.
- ❖ Oxides of elements in the center of the periodic table → **Amphoteric or neutral**.
- ❖ Most abundant elements on the earth surface → Oxygen (non-metal) > Silicon > Aluminium (metal).

Property	Definition	Across the Period (→)	Down the Group (↓)
Ionization enthalpy	Amount of energy required to remove electrons from the outermost or valence shell.	Increases	Decreases
Electron gain enthalpy	Amount of energy released in the process of adding an electron to an isolated gaseous atom. Therefore it is negative.	More negative	Less negative
Electronegativity	Tendency to attract shared pairs of electrons.	Increases	Decreases
Atomic radii	Distance between the outermost shell and the nucleus of the atom.	Decreases	Increases
Metallic character	Tendency to lose an electron.	Decreases	Increases
Melting Point	Temperature at which a substance in solid form converts into liquid.	Increases	Decreases

Metals, Minerals, Ores

❖ Principal Ores of some Important Metals :

Metals	Ores
1. Aluminium	a. Bauxite $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ b. Kaolinite $(\text{Al}_2(\text{OH})_4\text{Si}_2\text{O}_5)$
2. Iron	a. Haematite (Fe_2O_3) b. Magnetite (Fe_3O_4) c. Siderite (FeCO_3) d. Iron pyrites (FeS_2)
3. Copper	a. Copper pyrites (CuFeS_2) b. Malachite ($\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$) c. Cuprite (Cu_2O) d. Copper glance (Cu_2S)
4. Zinc	a. Zinc blende or Sphalerite (ZnS) b. Calamine (ZnCO_3) c. Zincite (ZnO)
5. Mercury	a. Cinnabar (HgS)
6. Tin	a. Cassiterite (SnO_2)
7. Lead	a. Galena (PbS)

- ❖ **Monazite** → Ore of Thorium, Lanthanum and Cerium.
- ❖ Bromine and Mercury → **liquid at room temperature.**
- ❖ Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) → used to make plaster of Paris ($\text{CaSO}_4 \cdot 1/2\text{H}_2\text{O}$)

Properties of Metals :

- ❖ Lustrous, Malleable, Ductile, Sonorous and good conductors of heat and electricity.
- ❖ Solids at room temperature, except **Mercury** which is a liquid.
- ❖ Metals combine with oxygen to form **basic oxides.**
- ❖ **Aluminium oxide** and **Zinc oxide** show the properties of both basic as well as

acidic oxides. These oxides are known as **Amphoteric oxides.**

- ❖ A more reactive metal displaces a less reactive metal from its salt solution.
- ❖ Metals occur in nature as free elements or in the form of their compounds.
- ❖ **Potassium** is the most reactive metal.
- ❖ Sapphire (Ruby) → Al_2O_3 .

Alloys

- ❖ An alloy is a homogeneous mixture of two or more metals, or a metal and a non-metal.
- ❖ In an alloy if one of the metals is mercury, then the alloy is known as an **amalgam.**
- ❖ The electrical conductivity and melting point of an alloy is less than that of pure metals.
- ❖ **Solder**, which is an alloy of lead and tin (**Pb** and **Sn**), has a low melting point. It is used for welding electrical wires together.
- ❖ Mercury does not form amalgam with iron.
- ❖ On mixing Iron with a small amount of carbon (about 0.05 %), it becomes hard and strong. And when iron is mixed with nickel and chromium, we get **stainless steel**, which is hard and does not rust.
- ❖ **Brass** - alloy of copper and zinc (Cu and Zn).
- ❖ **Bronze** - alloy of copper and tin (Cu and Sn).
- ❖ **Nichrome** - alloy of nickel and chromium, used in resistance wire.

Non - Metals

Properties of Non-metals:

- ❖ They are neither malleable nor ductile.
- ❖ They are bad conductors of heat and electricity, **except for graphite**, which conducts electricity.
- ❖ Non-metals form oxides which are either acidic or neutral.
- ❖ Allotropes of carbon:

Graphite	Diamond	Fullerene
❖ Hexagonal structure. ❖ Graphite is the thermodynamically most stable allotrope of carbon. ❖ It is soft and is used to prepare lead pencils. ❖ Graphite is used as a dry lubricant in machines running at high temperature, ❖ Graphite conducts electricity.	❖ Diamond has cubic structure (each carbon linked to another carbon atom in a Tetrahedral manner). ❖ It is the hardest substance on the earth. ❖ It is used as an abrasive for sharpening hard tools. ❖ Its thermal conductivity is highest with respect to other natural materials.	❖ The C-60 molecule has a shape like a soccer ball and is called Buckminsterfullerene. ❖ It is made of 20 hexagons and 12 pentagons. ❖ It has application in the drug delivery system in the body.

- ❖ **Solid (or frozen) carbon dioxide** → Dry ice, (as it looks like ice and changes directly from solid to gaseous form (sublimation)).
- ❖ It is used in preserving frozen foods where mechanical cooling is unavailable.

Types of Coal	% of Carbon Content
Lignite	25-35%
Sub Bituminous	35-45%
Bituminous	45-85%
Anthracite	> 85%.

Hydrogen and its Compounds

Hydrogen

- ❖ Symbol H
- ❖ Atomic number 1
- ❖ Electronic configuration 1s.
- ❖ Lightest element in the periodic table (atomic weight = 1.008u)

Isotopes of Hydrogen :

- Protium - ${}_1\text{H}^1$ (A = 1) no neutron – stable form
- Deuterium - ${}_1\text{H}^2$ (A = 2) one neutron – stable form
- Tritium - ${}_1\text{H}^3$ (A = 3) two neutrons - unstable and radioactive.

Water (H_2O):

- ❖ Boiling point of water: 100°C .
- ❖ Freezing point of water: 0°C .
- ❖ Density of water is maximum at 4°C .

- ❖ Thermal expansion of water helps in preserving the aquatic life in the water bodies during winters.

Heavy water (D_2O) :

- ❖ Oxide of deuterium.
- ❖ Used as moderator and coolant in nuclear reactors.

Soft water	Hard water
❖ Sodium is present as a cation. ❖ Do not contain dissolved salts of Ca or Mg. ❖ Easily forms froth with soap.	❖ contain dissolved salts of Ca or Mg and solubility of these salts determines the hardness of the water. ❖ Does not form froth with soap easily.

Hydrogen Peroxide :

- ❖ Chemical formula H_2O_2
- ❖ It is a weak acid.
- ❖ Used as an antiseptic, fuel, etc.

Some Important Elements and their Compound

- ❖ Zinc oxide is known as **flower of Zinc** or **Chinese white** & used in white paints
- ❖ **Silver nitrate** is used as a marker during election.

Sulphur

- ❖ Nature - Non-metal
- ❖ Color - Bright yellow at room temperature

- ❖ Form - Crystalline solid.
- ❖ Also known as '**Brimstone**' which means burning stone.
- ❖ Used in the production of H_2SO_4 and fertilizers.

Sulphur dioxide (SO_2)

- ❖ Toxic gas with a burnt match smell.
- ❖ Used as a precursor to H_2SO_4 , acts as a food preservative for dried fruits, used as a refrigerant, etc.
- ❖ An air pollutant.

Nitrogen gas (N_2)

- ❖ forms about 78% of the Earth's atmosphere.
- ❖ Found in amino acids, nucleic acids, hence support growth and reproduction in plants and animals.
- ❖ N_2 gas is used in filling aircraft tyres, as it does not easily react with other elements and also maintains tyre pressure for a long time as it does not diffuse through the rubber tyres easily.
- ❖ Insectivorous plants grow, where the soil is Nitrogen deficient.

Nitrous Oxide (N_2O):

- ❖ commonly known as laughing gas.
- ❖ Used as anesthesia in surgery and dentistry to reduce pain.

Ammonia (NH_3):

- ❖ colorless gas, with a pungent smell.
- ❖ Produced by Haber's process.
- ❖ Used as fertilizers.

Tear Gas (chloroacetophenone + Nitrochloroform):

- ❖ When enters our eye, it activates the nerves of the lachrymal gland to produce tears,
- ❖ eg- Alphachloroacetophenone, 2 - Chloro-benzalmalononitrile etc.

Halogens (Group 17):

- ❖ Elements of group 17 of the periodic table (fluorine, chlorine, bromine, iodine and astatine) are called Halogens.
- ❖ Halogens- Most reactive among nonmetals.
- ❖ Fluorine is the most reactive halogen.
- ❖ Fluorine, Chlorine-Gas.

- ❖ Iodine- Blue-Black Solid.
- ❖ Bromine- Liquid.
- ❖ Astatine- Radioactive.

Inert Gases (Group 18):

- ❖ Also known as Noble Gases due to extremely low reactivity with other elements.
- ❖ Mixture of Helium and Oxygen gas is used by the Scuba Divers to prevent Nitrogen Narcosis.

Acid, Base and Salt

Acid : It gives H^+ ion when dissolved in water.

- ❖ It is sour in taste.
- ❖ It turns Blue litmus Red .
- ❖ It is soluble in water.
- ❖ When an acid reacts with a metal, hydrogen gas is evolved and a corresponding salt is formed.
- ❖ An acid reacts with a metal carbonate or metal hydrogen carbonate, it gives corresponding salt, carbon dioxide gas and water.
- ❖ Vitamin C $\rightarrow$ Ascorbic acid.
- ❖ Oxalic acid is used in photography.

Substance	Acids present in them
Onion	Sulphonic acid
Apple	Maleic acid
Butter	Butyric acid
Eye-Drop	Boric acid
Cold-Drinks	Carbonic acid
Lemon, Orange	Citric acid
Grapes, Banana, Tamarind	Tartaric acid
Sour Milk	Lactic acid
Tomato	Oxalic acid
Vinegar	Acetic acid
Ant sting, Nettle sting	Methanoic acid

Base: It gives OH^- ion when dissolved in water.

- ❖ It is Bitter in taste, and have a slippery texture.
- ❖ Base turns Red litmus Blue.

- ❖ Most of the Bases are not soluble in water except **Alkali bases** (KOH, NaOH, NH₄OH).
- ❖ Most of the metals do not react with bases, but when some metals like **Aluminium** and **Zinc** reacts with bases, hydrogen gas is evolved with the formation of a complex salt which has a negative ion composed of the metal and oxygen.
- ❖ Milk of Magnesia [Mg (OH)₂] is an excellent antacid.
 - Mixing concentrated acids or bases with water is a **highly exothermic process**.
 - **Water of Crystallization** is the fixed number of water molecules present in one formula unit of a salt.
 - Eg: Hydrate Copper Sulphate (CuSO₄.5H₂O), Gypsum (CaSO₄.2H₂O), Plaster of Paris (CaSO₄.1/2H₂O).
 - Plaster of Paris is used by the doctors for supporting the fractured bones. It is also used in making toys, materials for decoration and for making a surface smooth.

Neutralization:

- ❖ Acids and Bases neutralize each other to form the corresponding salts and water.
- ❖ Acid + Base → Salt + H₂O.
- ❖ Eg: Bee sting (formic acid) + Baking soda (base) → reduction in pain (neutralization).

Salts:

- ❖ Salts of strong acid + strong base → Neutral (pH=7)
- ❖ Salts of strong acid + weak base → Acidic (pH<7)
- ❖ Salts of weak acid + strong base → Basic (pH>7)
- ❖ An aqueous solution of sodium-chloride is called **brine**.
- ❖ **Sodium hydroxide**→.

$$2\text{NaCl(aq)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{Cl}_2\text{(g)} + \text{H}_2\text{(g)}$$
 - The process is called the **chlor-alkali process** because of the products formed—'chlor' for chlorine and alkali for sodium hydroxide.

- ❖ **Bleaching powder** - Bleaching powder is produced by the action of chlorine on dry slaked lime [Ca(OH)₂]. Bleaching powder is represented as **CaOCl₂**.

- $\text{Ca(OH)}_2 + \text{Cl}_2 \rightarrow \text{CaOCl}_2 + \text{H}_2\text{O}$
- Bleaching powder is used – for bleaching cotton and linen in the textile industry, wood pulp in paper factories and for bleaching washed clothes in laundry.
- Used as an oxidizing agent in many chemical industries.
- To make drinking water free from germs.

- ❖ **Baking soda**→ The baking soda is commonly used in the kitchen for making cakes, *dhoklas* etc. Sometimes it is added for faster cooking. The chemical name of the compound is **Sodium hydrogen carbonate (NaHCO₃)**.

- **Uses of Baking soda**- For making baking powder, which is a mixture of baking soda(sodium hydrogen carbonate) and a mild edible acid such as tartaric acid. When baking powder is heated or mixed in water, Carbon dioxide produced during the reaction can cause bread or cake to rise, making them soft and spongy.
- Sodium hydrogen Carbonate is also an ingredient in **antacids**.
- It is also used in **Soda-acid fire extinguishers**.

- ❖ **Washing soda**→ Na₂CO₃.10H₂O (washing soda).

- It is also a basic salt.
- Uses of washing soda—Sodium carbonate (washing soda) is used in glass, soap and paper industries.
- It is used in the manufacture of sodium compounds such as borax.
- Sodium carbonate can be used as a cleaning agent for domestic purposes.
- It is used for removing permanent hardness of water.

pH scale:

- ❖ Strength of an acid or an alkali can be tested by using a scale called the pH scale (0-14).

- ❖ Measures the amount of hydrogen ion concentration in a solution.
- ❖ Neutral solution: $\text{pH} = 7$.
- ❖ Acidic solution: $\text{pH} < 7$.
- ❖ Basic solution : $\text{pH} > 7$.

Substance	pH value
Gastric juice	About 1.2
Lemon juice	About 2.2
Blood	7.4
Milk of magnesia	10
NaOH solution	10
Human body	7.0-7.8
Acid rain	< 5.6
Tooth decay	< 5.5

Indicators:

	Indicator	With Acid	With Base
Natural Indicators	Shoe flower(China Rose)	Magenta	Green
	Turmeric	No Change	Red
	Litmus	Red	Blue
Synthetic Indicators	Phenolphthalein	Remains Colourless	Pink
	Methyl orange	Red	Yellow

Acid Rain:

- ❖ If the pH of the rain water is **less than 5.6**, then that rain is called Acidic-rain.
- ❖ Composition of Acid Rain- Sulphuric acid, Nitric acid, Formic acid.

Sulphuricacid:

- ❖ Atmosphere of Venus is made up of thick white + yellow clouds of Sulphuric acids.
- ❖ Also known as **king of Acids, Oil of vitriol, Mattling Acid, Backbone of the chemical industry.**
- ❖ used in car batteries.

Aqua-Regia:

- ❖ Also known as **'royal water'**,
- ❖ Freshly prepared mixture of 3 moles of conc. HCl + 1 mole of conc. HNO_3 .

- ❖ It is a highly corrosive fuming liquid which can dissolve elements such as 'Gold' & 'Platinum'.

Nitroglycerin:

- ❖ Glycerine + Nitric Acid react in the presence of Sulphuric acid to form Nitroglycerine.
- ❖ It is called **'Noble Oil'**.

Soaps and Detergents

- ❖ Soaps are used for cleaning purposes, and are made up of sodium or potassium salts of long chain fatty acids.
- ❖ **Saponification** - Soaps containing sodium salts are formed by heating fat (i.e., glyceryl ester of fatty acid) with aqueous sodium hydroxide solution.

- ❖ Ester (vegetable oil) + alkali (NaOH, KOH) → salt (potassium or sodium salt of long chain carboxylic acid) + Glycerol (By product).
- ❖ Soaps do not work in hard water because hard water contains calcium and magnesium ions which form insoluble calcium and magnesium salts as precipitate, also known as **Scum**, when soaps are dissolved in hard water.

Soap fillers	Effect
Sodium Resinate	Foam.
Sodium silicate	Soap retains shape in water.
Glycerol	Provide Smoothness to skin
Sulphur	Helpful in Treatment of Acne

- ❖ Cleansing action of soap - Hydrophobic or non-ionic part of the soap gets attached to the dirt.
- ❖ **Detergents** : Sodium/Potassium salt of long chain benzene Sulphonic acid.
- ❖ Detergents get preference over soaps because they work even in hard water.
- ❖ Detergents with a straight chain of hydrocarbons are preferred over branched chains as the branched are non-biodegradable and consequently cause environmental pollution.

Polymers

- ❖ Natural Polymer – Silk, Cotton, Wool, Natural Rubber.
- ❖ Artificial Polymer – Polythene, Nylon, Teflon, Rayon (artificial silk).
- ❖ **Rayon** is obtained from a natural source, Wood pulp, yet it is a man-made fiber. It is cheaper than silk and can be woven like silk fibers. It can also be dyed in a wide variety of colors. Rayon is mixed with cotton to make bed sheets or mixed with wool to make carpets.
- ❖ **Bio-plastics** – biodegradable plastic made up of sugar present in the plant.
- ❖ **Vulcanization** of natural rubber - natural rubber is cross linked with Sulphur at a

temperature of 373-415 K to get vulcanized rubber which doesn't possess temperature sensitive properties of softness or brittleness like natural rubber.

- ❖ If Sulphur content is 5 % , then the rubber obtained is used in **tyres**.
- ❖ **Elastomers** : Rubber-like solids with elastic properties. Eg- buna-S, buna-N, neoprene.

Synthetic Rubbers

- ❖ **Neoprene** - It has superior resistance to vegetable and mineral oils. It is used for making conveyor belts, gaskets and hoses.
- ❖ **Buna – N** - It is resistant to the action of petrol, lubricating oil and organic solvents. It is used in making oil seals, tank lining, etc.
- ❖ **Fibers** are the thread forming solids which possess high tensile strength and high modulus. The examples are polyamides (nylon 6, 6), polyesters (terylene), etc.
- ❖ **Thermoplastic** - Its shape changes with change in temperature. Eg- PVC, polythene, poly-styrene.
- ❖ **Thermosetting plastic** - Its shape doesn't change with change in temperature. Eg - Bakelite, urea formaldehyde resins
- ❖ **Nylon** - It is the first synthetic fiber, discovered in 1931. Used in making clothes, socks, ropes, tents, toothbrushes, car seat belts, sleeping-bags, curtains, making parachutes and ropes for rock climbing.
- ❖ **Kevlar** is a material commonly used to make **bulletproof vests**.
- ❖ Biodegradable polymers → PHBV and Nylon-2- Nylon-6.

Chemistry in everyday life:

Food preservatives:

- ❖ Food preservatives prevent spoilage of food due to microbial growth and increase their shelf life. Eg: **table salt, sugar, vegetable oils and sodium benzoate**(C_6H_5COONa).
- ❖ **Sorbic-acid** or sulphites used to preserve jelly, pickles.

- ❖ **Sodium Benzoate** is used as a preservative in cold-drinks, pickles, jam, jelly, etc.
- ❖ **Nitrate/Nitrites** and **sodium tri-polyphosphates** are used in meat preservation.
- ❖ Candelilla wax, bee wax, sugarcane wax are in wax-coating on fruits.
- ❖ **Nitrogen** is used as a preservative in chips-packet.

Drugs and Medicines:

- ❖ **Analgesics:** Analgesics **reduce or abolish pain** without causing impairment of consciousness, mental confusion, in coordination or paralysis or some other disturbances of the nervous system. These are classified as follows:
 - (i) Non-narcotic (non-addictive) analgesics—Aspirin and Paracetamol (reduces fever (antipyretic) and prevents platelet coagulation. Because of its anti blood clotting action, aspirin finds use in prevention of heart attacks.
 - (ii) Narcotic drugs—Morphine (used for the relief of postoperative pain, cardiac pain and pains of terminal cancer, and in child birth).
- ❖ **Antibiotics:** Antibiotics are used as drugs to **treat infections** because of their low toxicity for humans and animals. **H.W. Florey** and **Alexander Fleming** shared the Nobel prize for Medicine in 1945 for their independent contributions to the development of penicillin.
- ❖ **Antiseptics:** Antiseptics are applied to the living tissues such as **wounds, cuts, ulcers and diseased skin surfaces**. Examples are **furacine, soframycin**, etc.
 - a. Dettol → mixture of chloroxylenol and terpineol.
 - b. Bithionol → added to soaps to impart antiseptic properties.

- c. Tincture of iodine → 2-3 percent iodine solution in alcohol-water mixture. It is applied on wounds.
 - d. Iodoform → used as an antiseptic for wounds.
 - e. Boric acid in dilute aqueous solution → weak antiseptic for eyes.
- ❖ **Disinfectants:** Disinfectants are applied to non-living objects such as **floors, drainage system, instruments, etc.**
 - **0.2 per cent solution of phenol is an antiseptic while its one percent solution is disinfectant.**
 - Chlorine in the concentration of 0.2 to 0.4 ppm in aqueous solution and sulphur dioxide in very low concentrations, are disinfectants.
 - ❖ **Antacids :** controls the over production of **acid in the stomach** which causes irritation and pain. eg: **sodium hydrogencarbonate, mixture of aluminium and magnesium hydroxide, cimetidine (tegamet), ranitidine (zantac)**, etc.
 - ❖ **Antihistamines:** used in the treatment of allergies. eg: **Brompheniramine (Dimetapp), terfenadine (Seldane)**.
 - ❖ **Tranquilizers:** Tranquilizers used for the **treatment of stress, anxiety, irritability, excitement, mild** or severe mental disease. Eg: **Noradrenaline** (mood change), **Iproniazid & phenelzine** (antidepressant), **chlordiazepoxide and meprobamate** (mild tranquilizers suitable for relieving tension) **Equanil** (used in controlling depression and hypertension).
 - ❖ **Anti-fertility drugs: To control population.**
 - ❖ **Breath Analyser kit** for alcohol: ($\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4$).

Artificial sweeteners:

- ❖ Ortho-sulfabenzamide, also called **saccharin**, is the **first popular artificial sweetening agent**.

- ❖ **Aspartame** is the most successful and widely used artificial sweetener. It is roughly **100 times as sweet as cane sugar**. Use of aspartame is limited to **cold foods and soft drinks** because it is **unstable at cooking temperature**.
- ❖ **Alitame** is a high potency sweetener, although it is **more stable than aspartame**, the control of sweetness of food is difficult while using it.
- ❖ **Sucralose** appears and tastes like sugar. It is **stable at cooking temperature**. It does not provide calories.

Miscellaneous

Aluminium	Produce silver and white flames and sparks.
Magnesium	Adds white sparks to fireworks.
Barium	Imparts green colors in fireworks.
Strontium salts	Responsible for red color to fireworks.
Calcium salts	Produce orange fireworks.
Cesium salts	Gives an indigo color in fireworks.
Copper	Imparts blue-green colors in fireworks
Lithium	Imparts a red color to fireworks.

- ❖ **Philosopher's wool - ZnO**.
It is an inorganic compound naturally found as a mineral Zincite. When exposed to air it reacts and leaves residue which appears wooly, so it is called "Philosopher's wool".
- ❖ **Neela Thotha**- $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (It is known as **Blue vitriol**)
- ❖ **Colloidal particles size** - (10^{-9} – 10^{-6} m).
- ❖ **Tyndall effect**- The scattering of a beam of light by colloidal particles is called the Tyndall effect.
- ❖ **White phosphorus** → used in rat-killer.
- ❖ Phosphorus is used in match-sticks in elemental form

- ❖ Phosphorus is also used to maintain the balance between the acidity of the blood and alkalinity of the blood.
- ❖ **Butane** → main component of LPG, lighter, etc.
- ❖ Colour of Black-Box in airplanes → generally orange.
- ❖ Graphite good conductor of heat and electricity.
- ❖ Diamond is a good conductor of heat, bad conductor of electricity.
- ❖ **Mica** - non-metal, good conductor of heat, but bad conductor of electricity, so used as an insulator.
- ❖ **Fool's gold** → Iron pyrite (FeS_2).
- ❖ Red colour of Ruby is due to **chromium**.
- ❖ Wrought iron is the purest form of Iron.
- ❖ Cooking at the mountains will be slower than the ground.
- ❖ Cooking in the pressure cooker is faster.
- ❖ Two pieces of metals stick to each other in space due to absence of air (vacuum) between them. This technique is used in **cold welding of metals**.
- ❖ Lead pipe is not used for drinking water, as it forms poisonous lead hydroxide.
- ❖ Yellow Phosphorus is kept in water & Sodium is kept in Kerosene oil.
- ❖ Ozone (O_3) is an allotrope of Oxygen.
- ❖ Artificial perfumes are prepared from Ethyl acetate.

Separation of substances:

- ❖ **Centrifugation**: Separation of mixtures through the use of Centrifugal force and spinning is called Centrifugation.
 - Used in diagnostic laboratories for blood and urine tests, in dairies and homes to separate butter from cream, in washing

machines to squeeze out water from wet clothes.

- ❖ **Distillation:** It is used for the separation of components of a mixture containing two miscible liquids that boil without decomposition and have sufficient difference in their boiling points.
- ❖ **Fractional Distillation:** To separate a mixture of two or more miscible liquids for which the difference in boiling points is **less than 25 K**, fractional distillation process is used, for example, the separation of different gases from air, different fractions from petroleum products, etc.

- ❖ **Crystallization** is a process that separates a pure solid in the form of its crystals from a solution. eg: Purification of salt that we get from sea water and Separation of crystals of alum (*phitkari*) from impure samples.
- ❖ **Sublimation (substance change directly from solid form to gaseous form)** process is used to separate substances such as **ammonium chloride, camphor, naphthalene** and **anthracene**.
- ❖ **Chromatography** is the technique used for separation of those solutes that dissolve in the same solvent. **Use of chromatography-** to separate colors in a dye, pigments from natural colors, drugs from blood.

